From Wallets to Wages: Consumer Income, Job Design, and Pay Disparities

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Abstract

Nearly half of US wage inequality occurs between firms. Prior research has traced between-firm pay differences to firm-level discretion in pay setting, or to differences in productivity and market power. Yet these firm-level factors are themselves tied to whom firms serve. Because higher-income consumers demand quality, firms serving them build jobs from higher-skill variants of similar tasks. And because workers who can perform many such tasks are scarce, these firms respond by specializing their jobs, not by enriching them. Matching consumer foot traffic data to administrative wage records, I find that establishments serving higher-income consumers pay their workers more, even comparing establishments within the same narrow industry and neighborhood. Analysis of online job postings shows that jobs at higher-income-serving firms bundle fewer tasks and, within similar categories of work, higher-priced variants of those tasks. Consumer inequality thus reaches inside firms, reorganizing the same occupations into different jobs and reemerging as pay inequality between firms in the service economy.

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1 Introduction

Workers performing similar jobs are often paid very differently depending on who employs them. These differences are not a marginal feature of the labor market: nearly half of overall wage inequality in the United States reflects variation in average pay between, rather than within, firms (Barth et al. 2016; Haltiwanger, Hyatt and Spletzer 2024). A large literature has sought to explain why some firms pay more than others, emphasizing organizational discretion in wage setting (Ton 2014) and differences in productivity and labor market power (Berger et al. 2024; Card et al. 2018). In contrast, I locate the source of the between-firm pay inequality outside the firm, in the consumers it serves. Because wages are paid out of what consumers spend, the incomes of a firm's customers constrain the quality it must deliver, the prices it can charge, and the jobs it creates. Job design is the channel through which inequality among consumers becomes pay inequality between firms. A fine-dining restaurant and a neighborhood diner both employ servers, yet they organize the same nominal occupation into substantively different jobs that command different pay. The restaurant's higher-income diners pay for quality, so it builds serving jobs from higher-standard variants of the same tasks. And because servers who can perform many tasks at that standard are scarce, it assigns those variants to narrow, dedicated roles rather than broad ones.

The account runs as follows. Firms align their offerings with the quality preferences of the consumers available to them, and because those preferences are strongly shaped by income, higher-income consumers demand higher-quality goods and services and are willing to pay a premium for them (Auer et al. 2024; Dai, Jin and Zou 2025). Delivering that quality requires changes to the organization of work. Firms serving wealthier consumers build jobs from tasks that support higher service standards (Boyle and Vandebroeck 2025; Sherman 2011). In a high-end restaurant, the job a casual restaurant gives to a single server – greeting guests, taking orders, setting tables – is split in two. Front servers provide personalized menu explanations and wine recommendations, while back servers maintain tables under strict protocols. Both halves carry service standards with little counterpart down-market. Because workers who can perform many such tasks at an elevated standard are scarce and expensive, bundling high-standard tasks into a single role commands a premium in the labor market; assigning them to dedicated specialists delivers the same quality at lower cost (Kohlhepp 2024), and the labor costs that remain can be passed on to consumers willing to pay for it. A lower-end restaurant faces the opposite calculus. Its customers demand fewer complex tasks, most of its workers can perform most of the tasks it requires, and broad assignments buffer the operational costs of high turnover (Atencio-De-Leon, Lee and Macaluso 2025). Job design thus links market position to pay. The task variants a firm selects determine the skills it must hire, and hence what it pays; how it bundles those variants into jobs determines how economically it can deliver that quality.

To test these predictions, I first construct a unique establishment-level dataset that captures both the income levels of consumers and the pay levels of employees, with a focus on customer-facing service industries. I build this dataset by linking restricted-use administrative records of average workplace pay from the Bureau of Labor Statistics with mobile location data on consumer visits. This is the first large-scale dataset to jointly measure these two variables at the firm level, allowing for a direct examination of the relationship between consumer income and worker pay rather than relying on occupation- or industry-level correlations in income and wage inequality (Gottlieb et al. 2023; Wilmers 2017). This linkage further separates the constraints imposed by consumers' incomes from firms' discretionary pay setting, an identification unavailable to prior case studies, which relied on price proxies or coarse market segment categories rather than direct income measures (Batt et al. 2020; Carré and Tilly 2020).

Analyses of this dataset show that establishments serving higher-income consumers pay more than others in the same neighborhood and narrow industry. In my most restrictive model, a 10% increase in consumer income is associated with a statistically significant 0.6% increase in average establishment pay. Because it compares establishments that share an industry and a neighborhood while serving different consumer bases, the design rules out mechanical geographic correlations arising from local cost-of-living differences or variation in the pools of customers and potential workers, factors that have long complicated efforts to identify the firm-level link between wages and product market segments (Batt et al. 2020). Additional robustness checks confirm that this relationship is not driven by tips or other forms of extra pay.

The predictive power of consumer income on pay is strongest in the accommodation and restaurant industries, where product markets are highly segmented by service quality or status concerns (Baum and Haveman 1997; Favaron, Di Stefano and Durand 2022). These differences are not a matter of occupational composition. Matching establishments to occupational employment data, I find that the same nominal occupation pays more where customers are richer, at magnitudes comparable to the establishment-level estimates, and most strongly among low-wage, frontline jobs that play an important role in delivering quality (Oliva and Sterman 2001; Sherman 2011). Moreover, I find evidence that wage differences across firms are shaped in part by within-establishment adjustments. Tracking individual establishments over time, I observe that firms raise pay as their customer bases shift toward higher incomes, with a 10% increase in consumer income corresponding to an approximate 0.1% rise in pay.

To investigate the mechanisms behind these patterns, I began by conducting informant interviews with restaurant managers across the Greater Boston area. These interviews highlighted differences in job design between establishments serving higher- versus lower-income consumer bases. To formally test the insights generated from the interviews, I turn to an online job postings dataset that includes detailed information on task requirements for each role. Comparing firms serving higher- versus lower-income consumers reveals systematic differences in job design within the same occupation. Firms serving higher-income consumers structure jobs around fewer distinct tasks, and within similar categories of work, they select the task variants that command higher prices in the labor market – differences concentrated in the task families characteristic of frontline work. Salary-posted listings suggest why jobs narrow. Task breadth commands a wage premium only at these firms, and only for high-standard task variants, so employers can contain the cost of quality by assigning high-standard tasks to narrow roles. I supplement this analysis with data from an online pay-review platform, which allows me to control for unobserved, time-invariant individual heterogeneity. The results point to worker selection rather than rent-sharing: firms serving richer consumers employ different workers, not the same workers at higher pay. Differences in job design thus reflect firms' efforts to deploy workers with different skills, rather than differences in practices applied to a common pool of workers.

This study advances understanding of between-firm pay inequality in three ways. First, it brings consumers into explanations of firm pay, identifying the consumer base as a market constraint that ties inequality among consumers to inequality among workers. Second, job design is the organizational mechanism behind this link: firms serving different consumer bases reorganize the same occupations into substantively different jobs. Third, contrary to the expectation from manufacturing research that firms enrich work to produce quality (MacDuffie 1995), those serving affluent consumers specialize jobs on average, because task breadth commands a premium where standards are high. Specialization, typically associated with low-end cost minimization, thus emerges in the customer-facing services studied here as the organizational form of quality at the high end of consumer markets. Together, the findings show how inequality outside the firm reaches inside it, reshaping jobs and generating pay differences across firms.

2 Explaining pay disparities between firms

2.1 Bringing Consumers into Explanations of Between-Firm Pay

While it has long been recognized that firms differ in how much they pay their workers (Abowd, Kramarz and Margolis 1999; Krueger and Summers 1988), recent attention to between-firm pay gaps has been fueled by mounting evidence that much of the rise in wage inequality over the past four decades has occurred between, rather than within, firms (Barth et al. 2016; Song et al. 2019). Research shows that some firms pay wage premiums and employ more skilled workers than others (Card et al. 2018; Song et al. 2019). But amid substantial work documenting these descriptive patterns, there has been less attention to why differences between firms exist in the first place. To sharpen the question, why would two competitors in the same industry, like Costco and Sam's Club, settle on paying their workers so differently?

My account runs from consumers, through job design, to pay. Higher-income consumers demand quality and are willing to pay for it. Firms serving them design jobs around tasks that support higher service standards, and performing those tasks requires skills that command higher prices in the labor market. Because workers who can execute many tasks at an elevated standard are scarce, these firms deliver quality by specializing jobs rather than broadening them, containing the cost of quality while securing its reliable production. Firms serving lower-income consumers face weaker quality demands and instead assign workers broad sets of general tasks that buffer the operational costs of turnover. On this account, a firm's product market position is a strategic choice, but one bounded by the pool of consumers available to it; conditional on the position taken, job design and pay follow as constrained complements of the niche the firm occupies (Adner, Csaszar and Zemsky 2014; Levinthal 1997). The account thereby supplies the organizational channel for a documented macro pattern: income inequality among consumers spills over into wage inequality (Gottlieb et al. 2023; Wilmers 2017); this spillover runs through which firms serve whom and what those firms pay.

Empirical research consistently finds that lower-income consumers are more price-sensitive, placing greater emphasis on price than on other product attributes (Auer et al. 2024; Dai, Jin and Zou 2025). It follows, then, that higher-income consumers tend to value non-price attributes – such as customization, service attentiveness, or symbolic value – more heavily (Gottlieb et al. 2023; Wilmers 2017), which I refer to broadly as *quality* throughout this paper. The well-established relationship between consumer income and quality preferences enables some firms to target more affluent customers by emphasizing higher-quality products and services as they seek to position themselves within distinct market niches (Podolny 1993; White 1981). Case studies of customer-facing establishments, for instance, show that firms serving higher-income consumers tend to offer superior service, reflecting the preferences of customers willing to pay a premium for quality (Batt et al. 2020; Carré and Tilly 2020; Talbert and Bose 1977).

How do firms respond to these demands for quality? Organizational research shows that firms structure jobs and allocate tasks to serve various goals – from cost control to expedient response to external regulations – resulting in substantial variation in task content even among similar roles

(Chan and Anteby 2016; Wilmers 2020). The demand for quality from higher-income consumers is another such goal. Firms serving these consumers assign workers distinct tasks to meet those elevated expectations. These differences often arise within similar categories of work: rather than taking on wholly different responsibilities, workers perform the quality-producing variants of functions common across an industry. For example, workers at grocery stores serving high-income customers custom-cut meats and operate scratch bakeries, whereas lower-end stores place less emphasis on customer interaction (Carré and Tilly 2020). Likewise, call center workers in higher value-added segments perform the same core task of handling customer calls, but rely less on scripted responses and dedicate more time to each customer (Batt 2000). In these instances, employers recognize that these higher-standard variants are essential for producing what they describe as "leading-edge goods and services" (Carré and Tilly 2020)."

Moreover, differences in job design may extend beyond a single task or role within a firm. When the quality of each task contributes critically to the overall quality of the final product or service (Kremer 1993), firms may find it advantageous to design jobs such that each worker performs the quality-producing variants of their tasks, in line with the expectations of high-income consumers, thereby raising skill requirements across the firm. For example, in a high-end restaurant, a lapse in customer interaction or cleanliness undermines the dining experience no matter how good the food is, so firms attend to the standard of every task rather than the food alone.

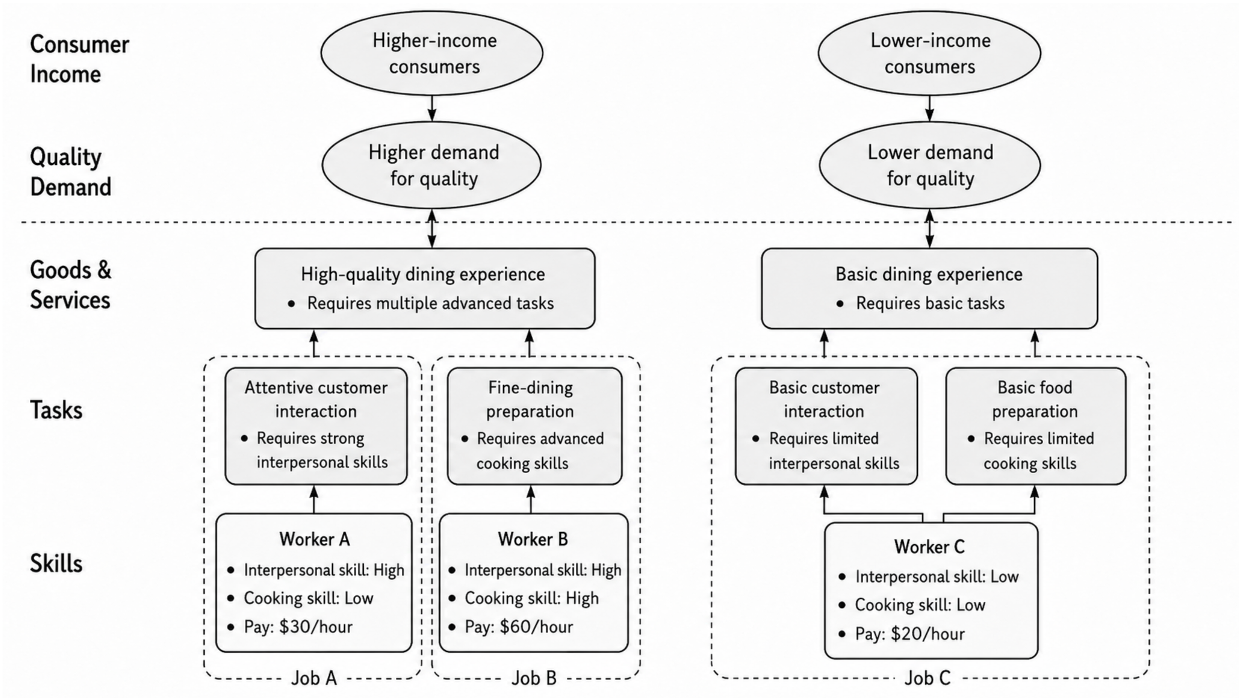
As firms require the performance of higher-standard tasks to serve quality-sensitive consumers, they will increasingly hire workers with the skills necessary to perform those tasks at the desired level of quality. Because richer consumers are willing to pay for these elevated quality levels (Gottlieb et al. 2023), market competition will, in turn, drive up compensation for the workers who possess those skills. For simplicity, I refer to these as *higher* skills: skills that command greater value in the labor market because they are necessary for performing quality-enhancing tasks. These may include interpersonal skills required for high-quality customer interactions, cognitive skills needed to remember and meticulously follow detailed service protocols, or other capabilities that support the production of higher-quality goods and services.

Importantly, firms can increase job specialization to ensure such advanced tasks are performed effectively, when workers who can perform each of those tasks are relatively scarce. At a baseline, firms may have an incentive to limit specialization to mitigate disruptions from turnover and to maintain workforce flexibility by shifting employees across different tasks (Atencio-De-Leon, Lee and Macaluso 2025). However, this approach becomes inefficient when the effective performance of advanced tasks is critical. In such cases, quality may suffer if a worker well-suited for one task is reassigned to another (Cici, Hendriock and Kempf 2025; Coraggio et al. 2025), while hiring employees who can perform multiple advanced tasks equally well can be excessively costly.[1] As a result, firms that serve high-income consumers are more likely to adopt specialized job structures, whereas their lower-end counterparts may prioritize flexible staffing strategies as a way to minimize the costs associated with turnover. Figure 1 summarizes the resulting framework.

Two families of existing explanations for between-firm pay differences serve as foils that sharpen this account. The first, rooted in organizational scholarship on employment, emphasizes firm-side discretion – administered pay systems (Cobb and Lin 2017; Doeringer and Piore 1971), intra-organizational bargaining (Avent-Holt and Tomaskovic-Devey 2014), and efficiency-wage (Krueger and Summers 1988) or good-jobs strategies (Rahmandad and Ton 2020; Ton 2014) – which share the implication that pay differences originate inside the firm, largely independent of the market

[1] This argument assumes that workers capable of meeting the demands of high-income consumers across multiple roles are relatively scarce compared to those who excel in specific roles (Kohlhepp 2024). Section 4.3.2 provides evidence consistent with this premise.

Figure 1: Theoretical framework and dining industry example



it serves. Yet critics argue that discretionary pay practices require rents that may not be available everywhere, least of all in customer-facing services, which combine intense price competition, low unionization, and a predominantly low-skilled workforce, leaving little room for firms to pay well by choice alone (Hall and Krueger 2012; Osterman 2018, 2020). Moreover, where some firms do pay well by choice, the question remains why they have rents to spend while competitors do not. The second emphasizes constraints rather than choices. With pay setting increasingly tied to external labor markets rather than internal administrative processes (Bidwell et al. 2013; Dencker and Fang 2016), a firm's pay is bounded by its position in those markets. Productivity differences shape which workers each firm can profitably employ, and wage-setting power lets firms pay even similar workers differently, which the economics literature formalizes as productivity-based sorting and monopsony (Berger et al. 2024; Card et al. 2018). These accounts discipline the discretion story yet treat the firm heterogeneity itself as a primitive: a firm's productivity and wage-setting power are taken as given, rather than as jointly determined with the market position it stakes out and the jobs it designs (Adner, Csaszar and Zemsky 2014; Levinthal 1997). If what a firm can charge constrains what it can pay and how it can organize work, then the rents that give some firms room for discretion and the firm heterogeneity that constraint accounts take as given are both tied to whom the firm serves. I test the account through three specific hypotheses, detailed below.

2.2 Consumer income and firm pay levels

First, I examine whether establishments that build their business models around and serve high-income consumers offer higher pay. While the notion that wage levels could be bound by the specific product market that firms operate in is not new (Commons 1909; Osterman 2018), prior research from different traditions has fallen short of clarifying how consumers' ability to pay and workers' pay levels are structurally linked at the firm level.

Hypothesis 1: Establishments targeting higher-income consumers offer higher pay.

Two streams of research address similar questions, and each stops short of the firm-level test the theory requires. These studies either (1) focus primarily on aggregate inequality at the occupation or industry level, rather than on how firms mediate the link between consumers and workers or (2) are limited in their ability to rule out major alternative explanations.

The first line of research examines how local income inequality among consumers influences wage inequality using large-scale quantitative datasets (Gottlieb et al. 2023; Wilmers 2017). These studies consistently suggest that greater income inequality translates into greater wage disparities.

However, the questions they address differ from the central question of this paper. While they investigate whether income inequality spills over into aggregate wage inequality, they provide limited insight into how firms mediate this relationship. The role of firms in this process is not straightforward: it could be that each firm occupies a market niche defined by the income level of its consumers and compensates workers accordingly, as posited here. Alternatively, a single firm may employ workers with varying pay levels who collectively serve consumers across different income groups. Because these studies rely on industry- or region-level data rather than firm-level observations, they are unable to distinguish between these mechanisms.

Another stream of research consists of case studies within specific industries or firms, which have consistently documented that firms occupying distinct product market positions compensate their workers differently. In particular, they have found that establishments in higher-end market segments or those with higher price points tend to offer higher wages (Batt et al. 2020; Carré and Tilly 2020; Talbert and Bose 1977).

Yet it remains unclear whether these studies successfully isolate the role of the consumer base in shaping pay. This ambiguity limits their theoretical implications. Existing studies often rely on speculative or self-reported categorizations of market segments, or on price-based proxies (Batt et al. 2020; Carré and Tilly 2020), rather than directly observing variation in the consumer bases served by different firms. As a result, these measures leave unclear whether the observed association between wages and product market position reflects consumer-driven market pressures or discretionary firm pay-setting decisions that are not fully explained by market incentives. For instance, some firms may choose to pay higher wages and charge higher prices by accepting lower profit margins, rather than calibrating wages to consumer income under product market competition. And because segments and price points are not denominated in income, the pay differences these studies document cannot be tied to the distribution of consumer income, leaving unexamined how inequality among consumers becomes pay inequality among workers.

Additionally, these studies often compare establishments cross-sectionally within broad geographic regions (Batt et al. 2020), making it difficult to disentangle wage differences driven by market segmentation from the regional correlation between income and wages. This issue is particularly pronounced in the United States, where high levels of neighborhood segregation contribute to substantial regional differences in consumer characteristics, even within the same metropolitan area (Berkes and Gaetani 2023; Boar and Giannone 2023). For example, firms located in wealthier neighborhoods may pay higher wages simply because the local cost of living is higher, regardless of the consumers they serve.

In this study, I analyze whether differences in consumer income within the product market predict pay levels across firms, addressing the limitations of prior research on the relationship between consumer characteristics and firm-level wage disparities. I theorize that consumer income levels – not merely a firm's intended product market positioning or a byproduct of regional factors – serve as a direct and positive predictor of establishment-level pay.

2.3 Consumer income and job design

Next, while prior research has established an initial link between firms' quality orientations and worker pay, the specific organizational mechanisms connecting the two remain less well understood. To explore such mechanisms in greater depth, I conducted informant interviews with restaurant owners and managers in the Greater Boston area – a consumer-facing service industry where establishments differ sharply in quality (Favaron, Di Stefano and Durand 2022) and where prior research has documented a strong association between market segments and pay levels (Batt et al. 2020). Although these interviews were conducted in the restaurant sector, they serve to motivate a more general mechanism applicable to consumer-facing service firms that compete on quality differences. The interviews revealed differences in job design across firms targeting different consumer segments, which informed the development of two hypotheses linking consumer income to job design in the service sector. Further details about the interviews are provided in Appendix G.

Organizational research has shown that the tasks comprising a job are not fixed (Cohen 2013; Wrzesniewski and Dutton 2001), and that significant heterogeneity exists in task composition even among otherwise similar jobs (Feldberg 2022; Martin-Caughey 2021). Existing studies highlight how job design, specifically through the allocation and recombination of tasks, is shaped by managerial responses to cost pressures (Wilmers 2020), organizational uncertainty (Miner 1987; Tan 2015), technological change (Beane 2023), and institutional environments (Lounsbury 2001).

This study introduces consumers – specifically their income levels – as an important but underexamined source of variation in task allocation across firms, with implications for pay differences. Consumers demonstrably leave an imprint on work in adjacent domains, redrawing professional jurisdictions (Bourmault and Anteby 2023) and reshaping worker identities (Chu 2024; DiBenigno 2022). Yet prior research on why job design and task allocation differ across firms has paid far less attention to consumers, despite this demonstrated influence on neighboring features of work. The closest antecedents are surveys of particular industries and occupations documenting specific task differences: whether service representatives work from scripts (Batt 2000), or whether machine operators also program their equipment when output is customized in small batches rather than mass-produced (Kelley 1990). Such studies indicate that task assignment varies with a firm's market context, but they capture particular tasks in particular settings rather than broader patterns in how jobs are composed, and the consumers behind those market contexts go unmeasured. I build on their lead by specifying which tasks should differ across consumer segments and why.

Firms serving higher-income consumers should design jobs around tasks that deliver the quality these consumers expect. Such quality-enhancing tasks often involve intensive customer interaction and personalized service (Batt 2000; Carré and Tilly 2020) – for example, "*in a fine dining restaurant, a server may spend 30-plus minutes tableside, talking about food, wine, the menu, and what's on the plate, and just having a conversation*" (Interviewee E). Alternatively, quality may be conveyed through tasks affecting the perceived value rather than functional performance, as in forms of "labor of distinction" (Boyle and Vandebroeck 2025) that work through status mechanisms. In other cases, quality rests on heightened precision in execution, formalized in protocols that specify exact service standards. As one owner of a higher-end restaurant put it, "*We have an SOP [standard operating procedure] for how to polish the tea cups, the teapot—we have an SOP for every detail*" required to "*consistently deliver the standard we set for our guests*" (Interviewee C). Regardless of the specific channel through which certain tasks contribute to the distinctive qualities valued by affluent consumers, workers with the skills to perform these tasks can command higher wages, as these consumers are willing to pay a premium for them.

Importantly, these quality-driven differences need not take the form of wholly different activities. They often operate within otherwise similar tasks: taking orders at a higher-end restaurant includes personalized recommendations and detailed menu explanations, whereas the same task at a lower-end restaurant involves a brief, transactional interaction. Elevated standards thus render tasks *de facto* distinct from their counterparts at lower-end firms. Within a similar category of work, jobs at high-end firms comprise the variants that produce quality, and the labor market prices these variants accordingly. I therefore expect the clearest signature of quality-driven job design to appear *within* comparable categories of work, in which variant of similar work a job performs. A job's overall task composition is a noisier signal because two forces reshape it at once. The demand for quality adds tasks that support higher standards, while the specialization dynamics described below narrow the set of tasks a job contains. Conditional on a job containing a given category of work, which variant it performs reflects the demand for quality alone.

> **Hypothesis 2**: Within similar categories of work, jobs at establishments targeting higher-income consumers comprise task variants that command higher prices in the labor market.

Job design responses to consumer demand extend beyond single tasks to how tasks are allocated across roles. Specifically, when workers who can perform multiple distinct tasks at consistently high standards are relatively scarce compared to those who excel at a single task, firms face strong incentives to increase job specialization in order to sustain quality. Although greater specialization may reduce the operational flexibility associated with broader job scopes (Atencio-De-Leon, Lee and Macaluso 2025), it enables firms to control labor costs while meeting elevated quality standards.

As one informant who oversees multiple restaurants across market segments explained: "*(In our high-end restaurant), we don't expect our servers or bartenders to participate in [polishing silverware] because they should be out with guests and selling*" (Interviewee E). This contrasts sharply with the more fluid job structure described by a manager at a casual restaurant: "*Only the general managers or those hired externally have set roles. Otherwise, staff here can easily jump between roles... If I'm bartending and something needs to be done management-wise, I'll do it. If I'm managing and they need help behind the bar, I'll do that too. Everyone just jumps in and helps each other out*" (Interviewee A). The same informant from the first quote further emphasized how job structures differ by market segment: "*[The restaurants have] different hierarchies... largely due to the increasing complexity of service at finer-dining vs. casual. This creates more positions* *[for finer-dining] on any given night of service and helps us delineate who is responsible for what without overloading anyone*" (Interviewee E). Supporting this view, Batt et al. (2020) finds that the proportion of workers performing both front-of-house and back-of-house roles is significantly higher in fast-food restaurants compared to higher-end establishments, highlighting how job specialization aligns with quality-driven market demands.

Prior employment research, largely rooted in manufacturing, emphasizes that firms can enhance product quality and workforce productivity by expanding workers' task variety and enriching job content, typically at the cost of higher pay (MacDuffie 1995; Rahmandad and Ton 2020). In service settings, however, firms on average make the opposite design choice: firms serving quality-sensitive consumers deliver quality by specializing jobs rather than enriching them. At the same time, organizational research has long documented the productivity gains of Taylorist forms of work organization, including their capacity to facilitate learning (Boh, Slaughter and Espinosa 2007; Narayanan, Balasubramanian and Swaminathan 2009) and reduce cognitive load (Gong and Png 2023; Staats and Gino 2012). Yet this research has paid less attention to why such systems become more or less prevalent under particular market conditions. Existing studies have largely attributed variation in specialization to inherent tradeoffs between the productivity benefits of specialization and task variety (Narayanan, Balasubramanian and Swaminathan 2009; Staats and Gino 2012), or to internal organizational factors such as coordination costs (Kohlhepp 2024). Firms serving quality-sensitive consumers instead have strong incentives to specialize jobs under labor market constraints. When workers capable of performing multiple high-level tasks are scarce, firms can reorganize work to ensure the reliable execution of critical tasks while limiting labor costs.

Hypothesis 3: Jobs at establishments targeting higher-income consumers comprise more specialized sets of tasks.

3 Predicting firm pay levels with customer income

To test the first hypothesis, I merge a mobile location dataset with administrative records on establishment-level employment and payroll, capturing consumer income and pay at the establishment level. The linkage enables the first large-scale analysis of the relationship between the incomes of the consumers a firm serves and the pay it offers.

3.1 Data

3.1.1 Consumer Income

The independent variable that will be used throughout this research, consumer income, is derived from the SafeGraph Patterns dataset. SafeGraph collects GPS-based mobile location data and provides monthly counts of unique visitors to each place of interest (POI), which primarily includes customer-facing businesses. The dataset covers the period from the first quarter of 2018 through the second quarter of 2022, which defines the temporal scope of the analysis. Because SafeGraph's coverage is relatively limited before 2020 and the years 2020-2021 were affected by the COVID-19 pandemic, I also replicate the main analysis using only the first two quarters of 2022 as a robustness check, reported in Appendix C.

In the SafeGraph dataset, visitor counts are aggregated by home census block group (CBG), enabling me to link each point of interest (POI) to the demographic and economic characteristics of its visitors' neighborhoods using 2016–2020 American Community Survey (ACS) data. I compute the average visitor income for each POI as a weighted average of CBG-level incomes, using visitor counts from each CBG as weights. Since ACS estimates reflect multi-year averages, CBG-level incomes are treated as fixed over time. However, the composition of visitors to each POI can vary from month to month, allowing for changes in a POI's average visitor income over time. The measure is reasonably reliable, as a CBG represents a small geographic unit, typically 600 to 3,000 residents. However, some within-CBG heterogeneity may still introduce measurement error. To address this concern, I re-estimate my main analysis using only the CBGs with more homogeneous income distributions as a robustness check, as detailed in Appendix C.

This visitor income measure does not exclusively capture *consumers*, as it may also include worker income to some extent. However, I expect the influence of workers to be minimal because each visitor is counted only once per month, regardless of how frequently they visit. Moreover, the average number of visitors in the dataset is significantly larger than the average employment size (see Table 1). Therefore, I anticipate that the variation in visitor income predominantly reflects consumer income.

3.1.2 Wage and Employment

I derive the workplace-level pay variable from the Quarterly Census of Employment and Wages (QCEW), a restricted-access administrative dataset provided by the Bureau of Labor Statistics. The QCEW dataset is sourced from state UI programs and covers all US establishments since 1990. Researchers have access to data from more than half of the states, which defines the scope of my final dataset. The list of included states is provided in Appendix A.

Importantly for this research, the QCEW contains monthly employment and quarterly wage data, allowing me to calculate monthly employment and pay per worker for each establishment.[2] The QCEW employment variable includes part-time and temporary workers, and the wage variable

[2]Note that the QCEW only include establishment-quarter-level wage averages. I supplement these records with a smaller survey that includes establishment-occupation wages in subsubsection 3.3.2.

includes bonuses, stock options, severance pay, profit-sharing, the cash value of meals and lodging, and tips and other gratuities.

While this broad definition of average pay is sufficient for documenting the relationship between consumer income and worker pay, it may obscure important mechanisms underlying this relationship – most notably the roles of tips and differences in hours worked – which the QCEW does not separately identify. To examine these mechanisms, I turn to an alternative data source in Table A.11 of Appendix L. Using pay review information from Glassdoor, I find no evidence that the main results are driven by tipping or by differences in reported work hours. Further details on the Glassdoor data and its linkage to SafeGraph are provided in Appendix K.

3.1.3 Merge Process

Both SafeGraph and the QCEW datasets include business names and address information, which I use to link workplace-level pay data to consumer incomes. The match rate – comparing the total pool of unique establishments in SafeGraph to those successfully matched – is approximately 36%, with a substantial portion of unmatched cases due to missing address information on the QCEW side. Details on the matching process and associated concerns are provided in Appendix B.

The final matched dataset used in my analysis comprises approximately ten million establishment-by-quarter pairs and around one million unique establishments. Table 1 presents descriptive statistics for this matched sample.

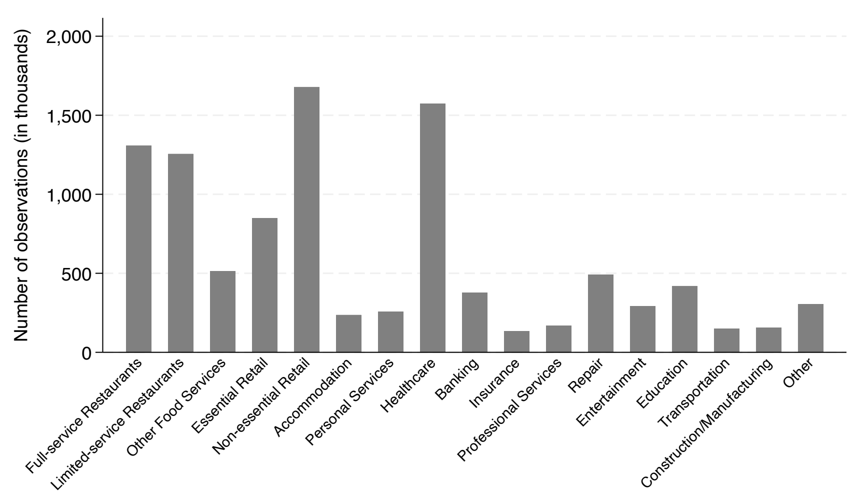
Figure 2 shows the industry composition of the final matched sample, which, as expected, is heavily concentrated in customer-facing sectors, reflecting the underlying structure of the SafeGraph dataset (Massenkoff and Wilmers 2023). Specifically, the sample largely consists of restaurants, retail, and healthcare establishments. However, the overrepresentation of the healthcare sector

Table 1: Descriptive Statistics of SG-QCEW matched sample (N=9,913,436)

Variable	Mean	S.D.	Min.	Max.
log(Average real pay)	7.63	0.67	-4.00	16.46
log(Median visitor income)	10.98	0.31	7.82	12.42
log(Average employment)	2.31	1.25	-1.09	10.40
Share Black visitors	0.09	0.12	0	1
Share White visitors	0.75	0.17	0	1
Share Hispanic visitors	0.17	0.18	0	1
Share Asian visitors	0.05	0.08	0	1
Share College graduates	0.31	0.13	0	1
Share Less than High school	0.11	0.07	0	1
Share Female	0.50	0.02	0	1
Median Age	39.24	4.58	7.5	86.5
log(Number of visitors)	7.55	1.62	1.81	15.26

* Observations at Time × Workplace level, 2018-2022.

Figure 2: Industry composition of the sample



Note: Data from SafeGraph-QCEW, 2018-2022.

could be a concern, as consumer income levels may be less relevant for this industry (Osterman 2020). I provide additional analysis excluding healthcare and manufacturing industry observations in Appendix C.

3.2 Methods

A key challenge in assessing the effect of consumer income on firm-level pay is disentangling it from geographic confounders – establishments located in higher-income areas tend to have both higher-income customers and higher-wage workers. The SafeGraph data help address this issue by providing detailed geographic information on both the home CBGs of visitors and the CBGs in which each POI is located. Using this information, I compare establishments within the same neighborhood and narrow industry, thereby ruling out geographic correlations between income and wages that arise from factors other than consumer demand, such as cost-of-living adjustments to pay. This design ensures that comparisons are made among establishments likely to draw from the same pools of customers and potential workers.

Specifically, I fit cross-sectional fixed effects models that control for quarter-by-CBG-by-industry fixed effects, as shown in Equation 1. This specification enables comparisons between establishments such as Dunkin' and Blue Bottle, or Walmart and Costco, that are located in the same neighborhood but draw consumers from neighborhoods with different income levels. For the industry variable, I use 6-digit codes from the North American Industry Classification System (NAICS), which identifies over a thousand distinct industries.

$$PAY_{et} = \beta * INC_{et} + \lambda \mathbf{X}_{et} + Time_t \times CBG_e \times NAICS_e \tag{1}$$

In Equation 1, PAY_{et} and INC_{et} represent the logged average pay of workers and the logged average income of visitors for establishment e at time t, respectively. Time is defined at the year-by-quarter level. $\mathbf{X}_{et}$ includes establishment-level controls such as the number of employees, the number of visitors, and the sociodemographic characteristics of the visitors.

I include controls for the number of employees and the number of visitors, as these variables may be correlated with wage levels through firm-size wage effects or overall revenue, and may also systematically relate to the income levels of consumers. For instance, although situated in a manufacturing context, prior research finds that larger firms tend to attract higher-income consumers (Faber and Fally 2022). Additional sociodemographic controls – such as age, race, gender, and education – are included in my most restrictive model given their well-documented associations with both income and consumption patterns (Jaravel 2019).

To test whether pay follows shifts in the consumers an establishment serves – the adaptation channel described in the theory – and to address the possibility that persistent unobserved differences across establishments, rather than consumer income itself, produce the cross-sectional association, I estimate a panel model that slightly modifies Equation 1.

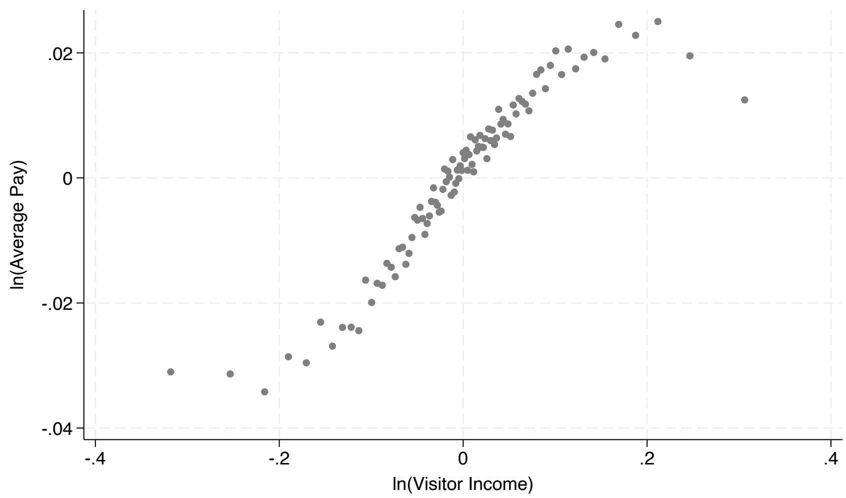
$$PAY_{et} = \beta * INC_{et} + \lambda \mathbf{X}_{et} + Time_t \times CBG_e \times NAICS_e + EST_e \tag{2}$$

Equation 2 incorporates workplace-specific fixed effects (EST_e), enabling within-establishment comparisons of worker pay and visitor income while accounting for temporal trends at the CBG × industry level. However, it is important to note that variation in visitor income over time within establishment is driven solely by compositional shifts in visitors' home CBGs, rather than reflecting changes in CBG-level income over time, as the income variable is derived from the 2016-2020 combined ACS.

3.3 Results

3.3.1 Does Visitor Income Predict Establishment Pay Levels?

Figure 3: Binned Scatterplot of Establishment Pay and Visitor Income, Adjusted for Time, Neighborhood, and Industry Effects



Note: Data from SafeGraph-QCEW, 2018-2022. Controlling for Time X CBG X NAICS. QCEW average pay is deflated with 2017 dollars. Visitor Income predicted with 2016-2020 combined ACS.

Figure 3 presents a binned scatterplot depicting the relationship between consumer income and worker pay levels. Each dot represents 1% of the visitor income distribution, residualized at the $Time \times CBG \times NAICS$ level. The plot reveals a clear upward-sloping linear trend, except at the very ends of the distribution, indicating that pay levels generally increase with visitor income across establishments.

Building on the graphical evidence presented in Figure 3, I next test whether the observed relationship holds under formal regression analyses with varying levels of control, reported in Table 2.

Table 2: Cross-sectional Pay Regressions on Visitor Income

	(1)	(2)	(3)	(4)
Visitor Income	0.227***	0.295***	0.101***	0.064***
	(0.001)	(0.001)	(0.002)	(0.004)
Fixed effects:				
Time × NAICS		×		
Time × NAICS × CBG			×	×
Controls				×
N	9,907,238	9,906,503	3,632,239	3,395,868

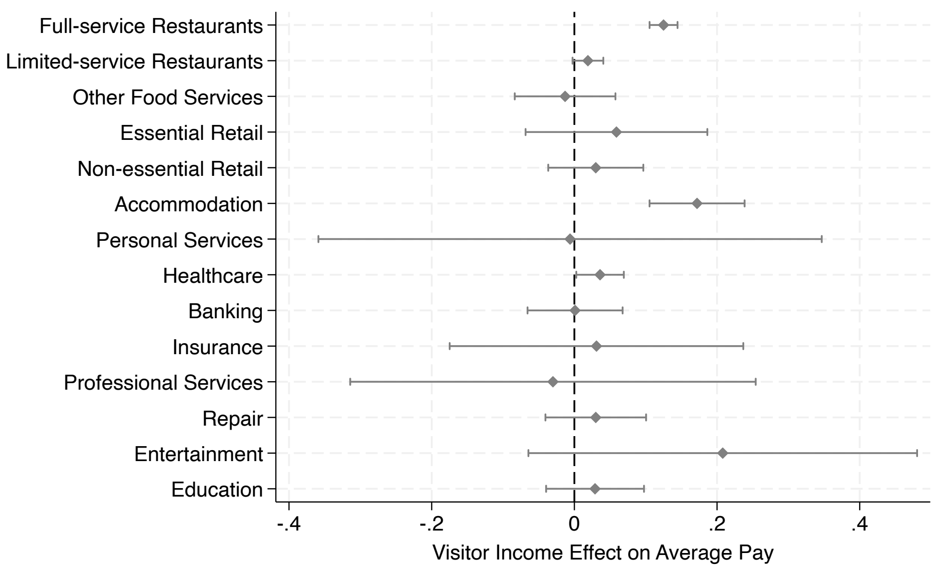
* Observations at Time × Workplace level, 2018-2022. Dependent variables for all models are logged average pay at establishment level from QCEW. Pay deflated with 2017 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Column (3) presents the coefficient from a specification that corresponds to the Figure 3, including fixed effects for time × CBG × industry. For comparison, Column (1) reports the unadjusted relationship with no fixed effects, while Column (2) includes time × industry fixed effects but omits CBG fixed effects.

The substantial decline in the coefficient from Column (2) to Column (3) indicates that approximately two-thirds of the cross-sectional correlation between visitor income and worker pay can be explained by the fact that high-income-serving establishments are disproportionately located in high-income neighborhoods. This implies that prior studies linking market segments to pay levels (*e.g.*, Batt et al. 2020) may have overstated the strength of this relationship by not including fine-grained geographic controls. The number of observations in Columns (3) and (4) drops to roughly one-third of the sample, which is expected, as including CBG fixed effects requires multiple establishments within the same narrow industry to be located in the same neighborhood.

Across all specifications, higher visitor income is consistently associated with higher average pay, lending support to the hypothesis that market segmentation by consumer income contributes to between-establishment pay differences. The most saturated model in Column (4), which follows from Equation 1, implies that a 10% increase in visitor income is associated with a 0.6% increase in worker pay. At the sample mean, this estimate suggests that a one-standard-deviation increase in visitor income corresponds to roughly $41 more in monthly pay per worker. Additional robustness checks in Appendix L suggest that this relationship is unlikely to be driven by extra pay, including tips, or by differences in work hours.

Figure 4: Coefficients by industry



Note: Data from SafeGraph-QCEW, 2018-2022. This figure illustrates the results from replicating the analysis in Column (4) of Table 2 by each industry sample.

Figure 4 breaks down the analysis from Column (4) of Table 2 by industry. The findings suggest that the overall positive relationship between visitor income and worker pay is primarily driven by restaurants and accommodation industries, which are known for high market segmentation based on quality or status differentiation at the establishment level (Baum and Haveman 1997; Favaron, Di Stefano and Durand 2022). This pattern is consistent with my theory that differences in customer preferences for quality may help explain how income inequality contributes to disparities in worker pay.

3.3.2 Occupational Analyses

While regressions of establishment-level pay on consumer income from Table 2 show that establishments serving higher-income consumers tend to pay their workers more on average, these models do not account for occupational composition. As a result, the observed pay gaps may reflect differences in the mix of occupations rather than wage differences within occupations. My theory of job design in response to consumer demand for quality does not rule out the possibility of occupational composition differences – establishments serving wealthier consumers may indeed employ a higher share of high-skilled roles, such as managers or first-line supervisors. However, if occupational composition alone accounts for most of the observed pay gap, it would weaken the empirical support for my framework, which also anticipates pay differences within occupations, not only between them. Such within-occupation differences may arise from cross-firm variation in quality expectations for workers with the same job titles, or from the presence of distinct roles, such as front and back servers in fine-dining restaurants, that are not differentiated by standard occupational codes from general servers in casual dining settings.

To distinguish between these two channels – occupational composition versus within-occupation pay differences – I merge in job-level employment and wage data from the Occupational Employment and Wage Statistics (OEWS) into the SafeGraph-QCEW matched workplace data. OEWS, a restricted-access dataset maintained by the Bureau of Labor Statistics, provides granular occupation-level wage and employment information for sampled establishments, spanning back to 1999. I link the two datasets using establishment identifiers common to both OEWS and QCEW. This allows me to extend the analyses in Table 2 by additionally controlling for occupations. Additional details on the OEWS dataset and its integration into this study are provided in Appendix E.

I begin by testing whether the higher pay associated with serving high-income consumers reflects differences in occupational composition or wage variation within occupations, as shown in Table A.4 and Figure A.3. Although the limited sample size from this linkage reduces statistical precision – widening standard errors and rendering the correlation between visitor income and wages statistically insignificant in most models – the stability of point estimates before and after controlling for occupation fixed effects in Figure A.3 provides suggestive evidence that consumer income is transmitted to worker pay primarily within occupations.

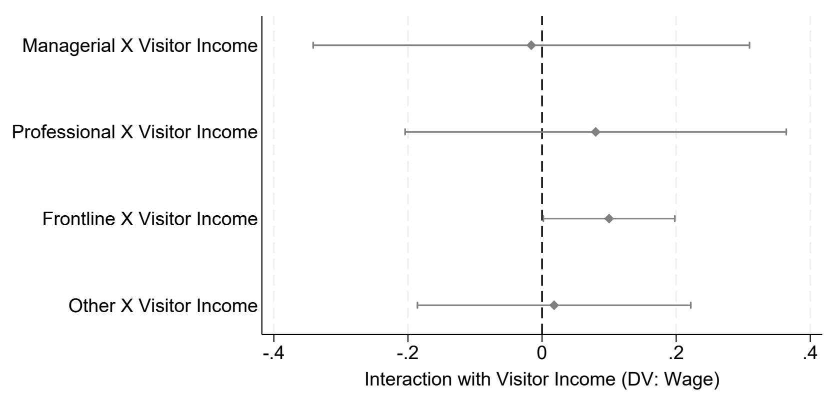
If the wage gains associated with serving high-income consumers occur within occupations, which specific occupations are most affected by consumer income levels? While my theory of job design does not predict that the mechanism operates exclusively through any particular occupation or task, it is premised on the idea that consumer demand for quality shapes job design and pay levels. If this is the case, we should observe within-occupation wage differences in roles that are known to be important to the production of quality. Conversely, an absence of such variation would call into question the proposed mechanism linking service quality to pay.

Previous research emphasizes that interaction with customers – whether direct or indirect – is a critical component in the production of service quality (Oliva and Sterman 2001; Sherman 2011). For this reason, I expect pay effects of consumer income among frontline occupations, which comprise a large share of the lower-wage workforce (Osterman 2020). To test this, I replicate the cross-sectional regressions of pay on visitor income with time × industry × CBG × occupation fixed effects (the specification with occupation fixed effects and controls in Figure A.3) across distinct occupational groups.

I first examine heterogeneity in the estimated effects across occupational pay terciles, defined using occupation fixed effects from a two-way workplace-occupation pay model following Wilmers and Aeppli (2021), estimated on the 2012 OEWS sample. The results in Table A.5, reported in Appendix F, reveal that the visitor income effect on wages operates primarily through the base category, the lower-paid occupations. Across model specifications, the consumer income effect on wages for lower-paid occupations was consistently larger than for other pay groups.

Next, I examine effect heterogeneity across occupations based on their roles and tasks. I categorized occupations into four groups using their 2-digit SOC codes: managers, professionals, frontline, and residuals as described in Appendix D.

Figure 5: Effect Heterogeneity by Different Types of Occupations



Note: Data from SafeGraph-QCEW-OEWS matched sample 2018-2022, dropping healthcare and manufacturing industries. Coefficients from fitting the specification with occupation fixed effects and controls in Figure A.3, separately for each subsample. The four types of occupations add up to the whole sample.

The results from Figure 5 suggest that the wage effect of consumer income is concentrated among frontline workers, those identified as directly involved in delivering service quality in previous studies (Oliva and Sterman 2001; Sherman 2011). Because the theory ties the pay premium to the quality demands on customer-facing work, frontline occupations are where the relationship should appear, and the estimates bear this out. The contrast across groups also weighs against the possibility that serving richer consumers raises pay across the board, which would yield comparable estimates for managers and professionals.

Taken together, the findings from occupational analyses suggest that wage gains associated with high-income consumers are not driven by differences in occupational composition, but rather by higher payoffs within occupations. Moreover, the relationship appears among lower-wage, frontline occupations, as the theory predicts. These patterns are consistent with the theory that pay differences reflect variation in product or service quality and highlight the importance of factors beyond occupational structure in explaining how pay varies across firms serving different consumer segments.

3.3.3 Within-establishment Changes Over Time

Does the relationship between consumer income and worker pay levels hold only across different establishments? The theory permits this possibility: when firms position at entry, job structures and pay arrive with the establishment and need not change afterward. A purely cross-sectional relationship, however, would be hard to distinguish from unobserved characteristics that are constant at the establishment level, even after controlling for narrowly defined industry and geographic cells. The results below show that, although firm pay levels are sticky over time, variation in consumer income within establishments predicts changes in pay, demonstrating the adaptation channel and showing that the association does not rest on fixed establishment characteristics alone.

Table 3: Panel Pay Regressions on Visitor Income

	(1)	(2)	(3)
Visitor Income	0.004***	0.007***	0.009***
	(0.001)	(0.002)	(0.003)
Fixed effects:			
Workplace	×	×	×
Time	×		
Time × NAICS × CBG		×	×
Controls			×
N	9,847,789	3,602,382	3,368,258

* Observations at Time × Workplace level, 2018-2022. Dependent variables for all models are logged average pay at establishment level from QCEW. Pay deflated with 2017 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. Standard errors clustered at each establishment level. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

The estimates from Equation 2 show that higher visitor income continues to predict higher worker pay within workplaces over time. A 10% increase in visitor income is associated with a statistically significant 0.1% increase in pay, with standard errors clustered at the workplace level. This pattern suggests that firms' pay practices evolve alongside shifts in the income composition of their consumer base. While the effect is substantially smaller than the cross-sectional relationship reported in Table 2, where a 10% increase in visitor income corresponds to a 0.6% increase in pay in the preferred specification, it shows that pay moves with changes in the consumer base itself, not only with fixed characteristics of the establishments serving it.

4 Predicting job design with customer income

This section examines the second and third hypotheses, which were formulated from qualitative insights derived from informant interviews with full-service restaurant managers in the Greater Boston area.[3] These hypotheses propose that jobs in establishments serving higher-income consumers comprise higher-standard variants of similar tasks and exhibit greater specialization. I formally test these propositions using task-level data from a large job postings dataset.

4.1 Data

The main dataset I use in this section is a matched dataset that I constructed by linking SafeGraph data with job postings from Indeed, collected and processed by Revelio Labs. The job postings, scraped by Revelio Labs starting in 2020, contain employer names and metro regions, which I leverage to link postings to establishments in the SafeGraph dataset. I restrict the sample to postings from 2022, the only year with full coverage from both sources (SafeGraph data extend only through 2022) and the year least likely to reflect distortions from the height of the COVID-19 pandemic. The sample also drops healthcare and manufacturing industries, consistent with the OEWS analyses in Section 3.3.2. In these industries, visitors are not the consumers who pay for output, as healthcare is financed largely by insurers rather than patients and manufacturing customers rarely visit the establishments that produce the goods, so visitor income does not measure the consumer base the theory concerns. The pay results do not depend on this restriction, as the main cross-sectional estimate is similar and slightly larger when these industries are excluded (Table A.2).

[3]As shown in Figure 4, the full-service restaurant industry is a major contributor to the observed correlation between consumer income and establishment-level pay. Further details on the interview methods, sample, and protocol are provided in Appendix G.

To classify the content of each posting, Revelio Labs applies an internal clustering algorithm that groups tasks based on textual similarity. The level of task granularity is determined by the number of clusters used in the algorithm (e.g., $k = 10, 50, 100, 200, 500, 1000, 1500$), where a higher k represents a more fine-grained breakdown of tasks. These clusters are hierarchical and nested by design. Each of the 50 task clusters falls exclusively within one of the 10 broader clusters, each of the 100 within one of the 50, and so on. I use these task classifications to construct both job design outcomes: the within-family price of the task variants a job comprises, which exploits this hierarchy, and job specialization, measured by the number of distinct tasks associated with each posting. In addition, a subset of postings discloses salary information, which I use to estimate the market prices of tasks.

Throughout this section, the unit of analysis is not the individual posting but a collapsed unit of postings, defined for each measure below. Collapsing is necessary because the consumer-base measure varies at the firm-by-metro level while postings number in the millions and are often near-duplicates within a firm. At the posting level, estimates would overstate precision and weight firms by their posting volume, letting a few high-volume chains dominate. Collapsing also keeps the units parallel to the previous section, which analyzes pay at the establishment level and, in the occupational analyses, at the establishment-by-occupation level. Because postings carry firm rather than establishment identifiers, the firm-by-metro unit stands in for the establishment, and the analyses below compare firm $\times$ metro $\times$ quarter $\times$ occupation units, or task families within them, rather than individual advertisements.

4.1.1 Task Prices and Variant Upgrading

My theory posits that, within similar categories of work, workers in firms targeting higher-income consumers perform higher-standard task variants that command higher pay. To measure the market value of tasks, I first infer task-specific prices by regressing the logged salary of each job posting on binary indicators for the tasks listed in the posting, estimated separately at each granularity used below ($k = 500$ or 1000). I estimate these regressions on the full universe of 2022 postings that report a salary and contain task information.[4]

I then construct two outcome measures from these prices. The first is the flat average: the mean price of the tasks listed in a job. The second exploits the nested structure of the task taxonomy to compare similar tasks across firms, following the theoretical expectation that quality differences materialize as higher-standard *variants* of similar activities. Using this hierarchy, I assign each narrow task to its broader task family and measure, within each family a job performs, the average price of the narrow variants chosen. The main specifications nest $k = 500$ variants within $K = 50$ families and $k = 1000$ variants within $K = 100$ families. In the analysis, per-posting values of the flat average are collapsed to firm × metro × quarter × occupation units, while the variant measure is computed at task-family cells within those units. Comparing this measure within family cells asks a sharper question than the flat average: when two jobs both involve, for example, retail customer service, does the job at the higher-income-serving firm perform the more highly priced variant of it? Within this $K = 100$ family, "Provide prompt, courteous service in retail environments" and "Create exceptional retail experiences to foster customer loyalty" are distinct $k = 1000$ variants describing different service standards, and the latter's inferred price is higher by roughly 15 percent of posted

[4]Approximately 11% of 2022 job postings in the Revelio Labs dataset include salary information. This aligns with prior research, which finds that only a small share of online job postings specify salaries – approximately 14% in the case of Batra, Michaud and Mongey (2023).

salary. The flat average, by contrast, mixes this variant margin with the composition of families a job spans, which specialization itself alters. For both measures, posted task descriptions likely understate true differences in execution standards, so estimated differences across firms should be read as conservative.

4.1.2 Specialization

Following prior research on job specialization (Boh, Slaughter and Espinosa 2007; Gong and Png 2023; Staats and Gino 2012), I measure specialization based on the number of distinct tasks assigned to workers in their roles (*i.e.*, task variety). Greater specialization corresponds to a lower number of tasks performed. To capture this, I construct three versions of the measure, counting distinct tasks at the $k = 100$, $k = 500$, and $k = 1000$ classification levels. Per-posting counts are averaged to the same firm × metro × quarter × occupation units.

While previous studies highlight an additional dimension of job specialization known as *job turf* – which reflects how distinct a worker's task profile is relative to their coworkers (Wilmers 2020) – I focus solely on task variety to assess specialization within firms. This choice reflects both theoretical and empirical considerations. The relationship between consumer income and job turf is likely to be more ambiguous than that with task variety. For example, firms serving richer consumers may expand staffing for certain specialized roles critical to service quality (e.g., hiring more front servers, thereby reducing job turf), while assigning specific responsibilities exclusively to a smaller subset of workers (e.g., creating host roles dedicated solely to greeting and seating customers, thereby increasing job turf). In addition, I am unable to measure job turf due to data limitations. Unlike prior research that relies on detailed information about internal task allocation within firms (Kohlhepp 2024; Wilmers 2020), my dataset is based on posted job descriptions and provides only partial visibility into actual task distributions.

Table 4 presents descriptive statistics for the core variables, computed at the collapsed units described above. All models include the logged number of visitors as a control to proxy for establishment size,[5] which may be correlated with both consumer segments and task allocation.

Table 4: Descriptive Statistics of SG-Job Postings matched sample

Variable	Mean	S.D.	Min.	Max.	N
Panel A: flat units (firm×metro×quarter×occupation)					
Visitor income (standardized)	-0.008	0.937	-2.075	1.683	150,051
ln(number of visitors)	8.282	1.736	1.811	15.145	150,051
ln(#distinct tasks), k=100	1.674	0.818	0.000	3.664	150,051
ln(#distinct tasks), k=500	1.865	0.907	0.000	4.263	150,051
ln(#distinct tasks), k=1000	1.892	0.923	0.000	4.394	150,051
Avg. task price, k=500 prices	-0.008	0.042	-0.622	0.331	150,051
Avg. task price, k=1000 prices	-0.011	0.044	-0.915	0.426	150,051
Panel B: nested units (additionally ×task family)					
Mean variant price, k=500 in K=50	-0.005	0.067	-0.622	0.375	1,047,677
visitor income (std.), same units	-0.020	0.922	-2.075	1.683	1,047,677
ln(number of visitors), same units	8.401	1.693	1.811	15.145	1,047,677
Mean variant price, k=1000 in K=100	-0.007	0.071	-0.915	0.430	1,254,462
visitor income (std.), same units	-0.010	0.920	-2.075	1.683	1,254,462
ln(number of visitors), same units	8.428	1.686	1.811	15.145	1,254,462

* Each observation is a collapsed unit of the matched sample, dropping healthcare and manufacturing industries. Panel A units enter the flat task price and specialization analyses; Panel B units enter the nested variant-price analyses. Unit values are unweighted means over the unit's postings. Visitor income standardized from deciles. Task prices are coefficients on task indicators from regressions of ln(posted salary) on the tasks listed in each posting, estimated on the full universe of 2022 posted-salary postings. Mean variant prices are the average prices of the narrow task variants a unit performs within each task family, with k=500 variants nested in K=50 families and k=1000 variants in K=100 families.

4.2 Methods

Unlike the SafeGraph dataset, which contains establishment-level identifiers, the job postings dataset includes only firm-level identifiers, and the most detailed geographic information available is at the

[5]Employment size data is unavailable in both the SafeGraph and job postings datasets.

metropolitan statistical area (MSA) level. Accordingly, I define employers at the firm-by-metro-region level. To construct consumer characteristics at this same level, I aggregate SafeGraph establishment data by averaging visitor income deciles across establishments, controlling for time × industry × metro region cells.[6] For the analyses below, I use this average-decile measure in continuous form, standardized across firm-by-metro units, so that coefficients are per standard deviation of the consumer-income measure. Further details on the procedure matching postings to SafeGraph establishments are provided in Appendix H.

As noted above, posting-level estimation would overstate precision and weight firms by their posting volume. The weighting is not innocuous: because chains standardize job design nationally, posting-weighted estimates understate how the typical firm's job design covaries with its clientele. Accordingly, I collapse postings to firm × metro × quarter × occupation cells (for the nested variant measure, additionally by task family) and estimate variants of Equation 3:

$$TASK_{fmot} = \beta INC_{fm} + \lambda \mathbf{X}_{fmt} + Time_t \times MSA_m \times NAICS_f \times SOC_o \tag{3}$$

In Equation 3, $TASK_{fmot}$ represents a job design outcome constructed from posted task content: the flat average task price or the logged number of distinct tasks. For the within-family variant price, the outcome instead varies at task-family cells within these units ($TASK_{fmotg}$ for family g). INC_{fm} is the standardized consumer-income measure defined above, and $\mathbf{X}_{fmt}$ includes the logged number of visitors to the firm's SafeGraph establishments in the metro region, proxying for establishment

[6] I use income deciles rather than raw consumer income when collapsing establishment-level data to the firm-by-metro level. Deciles are computed from establishment-level visitor income residualized on metro × industry × quarter cells, so that an establishment's value reflects its position relative to its local market rather than regional differences in income levels. Ranking the residuals into deciles then bounds the influence of establishments located in extreme-income neighborhoods when averaging across a firm's establishments (with visitor weights), and places all firms on a common scale.

size. The fixed effects compare cells within the same quarter, metropolitan area, 6-digit NAICS industry, and 6-digit SOC occupation (and, for the nested measure, the same task family), so that β captures differences in how firms serving different consumer bases design jobs for the same occupation, posted in the same market at the same time – with each firm's presence in a market weighted equally rather than by posting volume. Occupation fixed effects are included both because the pay analyses already locate the consumer-income relationship within occupations (Figure A.3) and because postings measure hiring flows rather than employment stocks, so that across-occupation comparisons would confound job architecture with firms' posting propensities. Standard errors are clustered by firm × metro, the level at which the consumer-base measure varies, throughout.

In Appendix I, I report a further robustness check: restricting the sample to firm-by-MSA observations for which the SafeGraph data indicate that the firm operated only one establishment in the MSA, making the match more likely to approximate an establishment-level match. If anything, the specialization results strengthen in this subsample, consistent with chains standardizing job design nationally, and the within-family variant estimate is likewise robust.

4.3 Results

4.3.1 Consumer Income and Task Content: Variant Upgrading

I begin by examining whether, within similar categories of work, higher consumer income is associated with workers performing task variants that command greater market value, testing Hypothesis 2.

Table 5 reports the results. Columns (1)–(2) use the flat average task price of the job, computed from $k = 500$ and $k = 1000$ prices respectively. The association with consumer income is positive

Table 5: Consumer Income and Task Prices

Dependent variable	Flat average		Variant price within task family (nested)			
Task clusters	k=500	k=1000	k=500-in-K=50		k=1000-in-K=100	
	(1)	(2)	(3)	(4)	(5)	(6)
Visitor income (std.)	0.0209***	0.0328***	0.0074**	0.0130***	0.0096***	0.0139***
	(0.0080)	(0.0080)	(0.0033)	(0.0049)	(0.0032)	(0.0042)
Fixed effects:						
Qtr. × MSA × Ind. × Occ.	×	×	×	×	×	×
× Task family			×	×	×	×
Controls	×	×	×	×	×	×
Sample	Full	Full	Full	NAICS 72	Full	NAICS 72
N	51,221	51,221	194,146	94,077	202,660	99,134

* Estimates in columns (1)-(2) are at firm × metro × quarter × occupation units, and in columns (3)-(6) at task-family cells within those units. The sample drops healthcare and manufacturing industries. Visitor income standardized across firm × metro units. Controls include the logged number of visitors. Task prices from the full-universe 2022 salary regression. Standard errors clustered by firm × metro in parentheses. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

economy-wide, meaning that jobs at firms serving higher-income consumers comprise higher-priced tasks on average, without distinguishing whether this reflects the task families a job spans or the variants performed within them. Columns (3)–(6) turn to the within-family variant measure, the direct test of Hypothesis 2. Comparing jobs within the same task family, occupation, market, and quarter, firms serving higher-income consumers perform variants that carry significantly higher prices, and the estimate is stable across the two nestings (columns 3 and 5). Columns (4) and (6) estimate the same specifications within accommodation and food services (NAICS 72), which I report separately here and in the analyses that follow because the sector drives much of the association between consumer income and worker pay (Figure 4). Within the sector, the estimates are larger across both nestings, so variant upgrading is strongest in the same sector that drives the association between consumer income and worker pay, as a task-content mechanism would predict. Because the design compares cells within the same task family and 6-digit occupation, the estimates isolate the margin Hypothesis 2 predicts. Nominally identical jobs, such as server positions at high-end and casual restaurants, are built from variants of similar activities, and firms serving higher-income consumers assign the higher-priced variants. In magnitude, a one-standard-deviation increase in consumer income shifts the average price of the variants a job performs by about 1 percent of posted salary within the same task family, and by about 1.4 percent within accommodation and food services. The within-family estimate is robust to restricting the sample to single-establishment firms, where it is, if anything, larger (Table A.8).

Which types of tasks carry the variant upgrading? Similar to the motivation for my occupation-level analysis in Section 3.3.2, while my theory of job design does not predict that the mechanism operates exclusively through any particular task, it is premised on the idea that consumer demand for quality shapes job design and pay levels. A clear implication is that upgrading should appear in tasks pivotal to delivering quality; its absence there would weaken the argument that consumer demand for quality links income, job design, and pay. Prior work identifies customer contact – direct or indirect – as central to producing service quality (Oliva and Sterman 2001; Sherman 2011), so I examine whether it appears in task families involving direct customer interaction and, more broadly, in those characteristic of frontline service work, whose tasks shape the customer experience directly or indirectly.

I classify the $K = 100$ task families along both dimensions. *Frontline* families are identified empirically from the occupational composition of the postings in which they appear, using the frontline-occupation definitions in Appendix D. *Customer-facing* families are classified from the substantive content of each family using a large language model (Claude Fable 5), with borderline cases reviewed manually; Appendix J details the classification procedure and lists the classification of each family.

Table 6: Within-Family Variant Upgrading by Task Family Type

	Frontline	Non-front.	Cust.-facing	Non-CF
All industries: income (std.)	0.0116***	0.0039	0.0104**	0.0086**
	(0.0036)	(0.0071)	(0.0051)	(0.0040)
NAICS 72: income (std.)	0.0160***	0.0042	0.0199***	0.0056
	(0.0048)	(0.0095)	(0.0064)	(0.0047)
Fixed effects:				
Qtr. × MSA × Ind. × Occ.	×	×	×	×
× Task family	×	×	×	×
Controls	×	×	×	×
N (All industries)	146,217	56,443	117,270	85,390
N (NAICS 72)	76,273	22,861	54,701	44,433

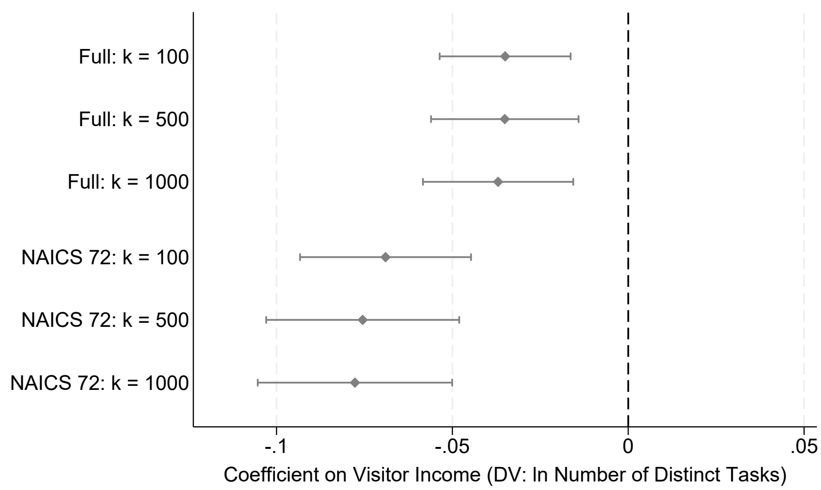
* All columns use the nested k=1000-in-K=100 specification, with estimates at firm × metro × quarter × occupation × task family cells. The sample drops healthcare and manufacturing industries. Appendix J lists the frontline and customer-facing classification of each task family. Visitor income standardized across firm × metro units. Controls include the logged number of visitors. Task prices from the full-universe 2022 salary regression. Standard errors clustered by firm × metro in parentheses.
* Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Table 6 estimates the within-family variant specification separately by task-family type. Variant upgrading is concentrated in frontline task families: the estimate among frontline families is three times that among non-frontline families in the pooled sample, and four times within accommodation and food services, a split that is robust across classification variants. The customer-facing split points the same way within accommodation and food services, while in the pooled sample both groups show positive estimates. Overall, upgrading appears in the tasks that shape the customer experience, as the quality mechanism implies, and both splits are sharpest within accommodation and food services, the sector where consumer income is most strongly tied to worker pay. This task-level pattern mirrors the occupational pattern in Figure 5: the frontline work in which the wage relationship concentrates is also where the higher-priced variants are performed.

4.3.2 Consumer Income and Job Specialization

Next, I test whether higher consumer income is associated with greater job specialization, measured by the number of distinct tasks associated with each job. This serves as a direct test of Hypothesis 3.

Figure 6: Consumer Income and Job Specialization

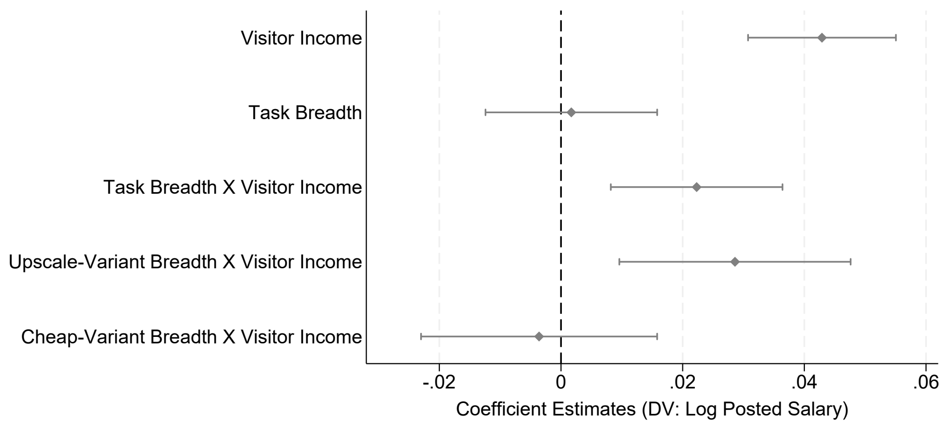


Note: Data from SafeGraph matched with Revelio job postings, 2022, dropping healthcare and manufacturing industries. Coefficients on standardized visitor income from regressions of the logged number of distinct tasks at the indicated task classification level. Estimates compare firm × metro × quarter × occupation units within quarter × MSA × industry × occupation cells, controlling for the logged number of visitors. Full estimates in Table A.7.

Figure 6 reports the results across three task classification schemes ($k = 100$, $k = 500$, and $k = 1000$), with the full estimates in Table A.7. Jobs at firms serving higher-income consumers consistently involve fewer distinct tasks: approximately 3.5 to 3.7 percent fewer per standard deviation of visitor income, within the same occupation, industry, market, and quarter. The association is again concentrated in accommodation and food services, the sector that drives the relationship between consumer income and worker pay. It reaches roughly 7 to 8 percent per standard deviation within the sector. The specialization estimates are also robust to the single-establishment restriction (Table A.8).

Why should high-clientele firms specialize? My theory holds that breadth becomes costly precisely where task standards are high: workers who can perform many tasks at an elevated standard are scarce, so bundling high-standard tasks into one role commands a premium in the labor market. The subsample of postings that disclose salaries allows a direct test of this premise.

Figure 7: The Price of Task Breadth by Consumer Segment



Note: Data from SafeGraph matched with Revelio job postings, 2022, dropping healthcare and manufacturing industries and restricted to postings that disclose salaries. Coefficients from regressions of log posted salary on standardized visitor income, task breadth, and the interaction of the two, where breadth is the logged number of distinct k=1000 tasks in the job. The top three estimates are from the pooled specification with overall breadth; the bottom two split breadth into upscale and cheap variants at the within-family median price and enter both interactions jointly. Estimates compare firm × metro × quarter × occupation units within quarter × MSA × industry × occupation cells, controlling for the logged number of visitors.

Figure 7 plots the central estimates from regressions of log posted salary on the consumer-income measure and the task structure of the job. The top estimate establishes the postings-side counterpart of Hypothesis 1, with posted salaries rising in visitor income. Task breadth carries no price on average. Breadth is priced *only* at high-clientele firms: bundling more tasks into a role commands a salary premium where the firm's consumer income is higher. Splitting each job's tasks into upscale and cheap variants at the within-family median price, the premium loads entirely on the number of *upscale* task variants a role spans, while spanning more cheap variants carries no premium. What is expensive for a high-clientele employer, in other words, is a role that spans many tasks in their high-standard variants, giving these firms an incentive to define roles narrowly, the logic restaurant managers articulated in the interviews quoted in Section 2.3.

Notably, specialization itself is not what commands the premium. At high-clientele firms broader roles command more pay, not less, so the consumer-income premium cannot reflect a return to assigning fewer tasks. The estimates thus establish two facts: bundling upscale variants is priced, and firms serving higher-income consumers assign narrower jobs, spanning fewer tasks on average (Figure 6). The theory advanced here reads the price as evidence that workers who can perform multiple high-standard tasks are scarce, and the narrowing as employers' response to contain this cost, with each high-standard task performed by a dedicated worker rather than by workers who could perform them all. Containment is partial, however. Firms serving higher-income consumers still pay more on average, because delivering quality requires performing higher-standard tasks and hiring the workers who can perform them. Specialization moderates the cost of quality but does not eliminate it. On this interpretation, specialization is the modal strategy by which employers in the customer-facing services studied here reliably deliver the quality that higher-income consumers demand while keeping its cost in check.

Taken together, the two job design results point in the same direction. Establishments serving higher-income consumers assign narrower jobs and, within the task families their jobs retain, perform the higher-priced variants (Table 5), and both margins concentrate in accommodation and food services, the sector where consumer income is most strongly tied to worker pay. On average, jobs at these establishments are thus not broader versions of ordinary roles but narrower roles built from upgraded tasks. Pay tracks the priced variants a job contains (Hypothesis 2), while specialization (Hypothesis 3) is the employer's device for containing their cost.

4.3.3 Worker Selection

The cost-containment account in the previous section presumes selection on skill: upgraded tasks are performed by more skilled workers, whom firms serving higher-income consumers hire at higher pay. Yet the pay premium could instead reflect rent-sharing mechanisms, such as compensation for greater effort (Cappelli and Neumark 2001; Cappelli and Chauvin 1991). That is, firms may not need to hire more skilled workers to perform the higher-priced task variants; instead, they could assign these more demanding tasks to a largely similar pool of workers by offering higher wages.

Assessing this alternative explanation is central to my theoretical interpretation of what differential job designs across firms serving different consumer groups represent. If higher pay reflects rent-sharing rather than selection on skill, observed specialization would not constitute a necessary response to skill scarcity. Instead, it could emerge as (1) a mechanical consequence of higher wages increasing worker attachment (Shapiro and Stiglitz 1984), thereby weakening employers' incentives for flexible task assignment (Atencio-De-Leon, Lee and Macaluso 2025), or (2) a means of accelerating learning-by-doing (Narayanan, Balasubramanian and Swaminathan 2009). Both mechanisms provide key alternative explanations to the theory advanced in this study.

To assess whether rent sharing is the primary driver of pay premiums associated with serving high-income consumers, I use a dataset that enables comparisons of pay outcomes for the same individual across different jobs. This allows me to distinguish whether pay differences are due to worker selection – where higher-skilled individuals sort into higher-paying firms – or to rent-sharing practices. Specifically, I match SafeGraph's visitor income data to wage reports from Glassdoor and leverage Glassdoor's anonymized user identifiers. Additional details on the Glassdoor dataset and the matching process are provided in Appendix K.

If pay premiums are primarily driven by firm-level rent sharing rather than differences in worker characteristics, then controlling for individual-specific factors should not substantially change the observed relationship between consumer income and pay. Conversely, a significant reduction in the coefficients would suggest that pay premiums in firms serving higher-income customers largely reflect worker selection effects. To test this, I estimate regressions of reported hourly pay on SafeGraph's high-income consumer indicator, both with and without individual user fixed effects.

Table 7: Worker Selection: Individual Fixed Effects

	(1)	(2)	(3)	(4)
High-inc. Consumer	0.0251**	0.0046	0.0167	0.0044
	(0.0114)	(0.0084)	(0.0104)	(0.0084)
Fixed effects:				
Yr. × City × NAICS	×	×	×	×
Individual		×		×
Controls	×	×	×	×
Yrs. experience			×	×
N	143,434	143,434	143,428	143,428

* Glassdoor pay reviews matched to SafeGraph, 2006–2023, restricted to reviews from users who submitted multiple reviews. The sample drops healthcare and manufacturing industries. The dependent variable is logged hourly pay. The high-income consumer indicator equals one when the firm-city's average visitor-income decile, computed within year × industry × city cells and averaged over time, exceeds 5.5 (Appendix K). All columns include year × industry × city fixed effects and control for logged visitor counts, logged firm size, employer type, and employment status; columns (3)–(4) add flexible controls for years of relevant experience. Standard errors clustered by employer × city in parentheses. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

The results in Table 7 suggest that pay premiums at firms serving higher-income consumers are largely attributable to differences in the workers they employ. Controlling for individual worker fixed effects reduces the estimated coefficient on the high-income consumer indicator by roughly 80 percent; adding flexible controls for years of relevant experience yields the same picture, with a 73 percent reduction. This attenuation challenges the view that pay premiums are primarily driven by rent-sharing and instead points to worker selection as an important contributor, lending support to my argument that employers design specialized jobs to minimize hiring costs in the labor market.

5 Discussion and Conclusion

This study asks how the incomes of the consumers a firm serves constrain the pay and design of its jobs in the service economy. The findings show that, even within the same narrow industry and neighborhood, consumer income is an important predictor of variation in average pay across establishments. The findings also indicate that pay levels move with shifts in the customer base within establishments over time. Robustness checks suggest that the results are unlikely to be driven solely by differences in work hours or tips and gratuities (Appendix L). Moreover, these pay differences are not primarily explained by differences in occupational composition across firms; rather, they emerge within occupations across firms, particularly among low-wage, frontline roles. In line with high-income consumers' demand for higher-quality goods and services, workers at the establishments serving them perform the higher-priced variants even within similar categories of work. The upgrading appears in the task families that shape the customer experience, where prior research locates the production of service quality (Oliva and Sterman 2001), and the relationship is strongest in accommodation and food services, a sector sharply segmented by quality and status (Baum and Haveman 1997; Favaron, Di Stefano and Durand 2022). Both patterns support a quality-based reading of the relationship.

In addition, firms specialize jobs to allocate tasks in ways that deliver quality reliably. When workers capable of performing multiple high-standard tasks are scarce, assigning them to duties outside their core expertise can compromise quality, while hiring individuals who can excel across multiple high-standard tasks can be prohibitively costly. In such contexts, job specialization becomes a more efficient strategy for ensuring service quality. The salaries posted in job advertisements bear this premise out directly: task breadth commands a wage premium only at firms serving high-income clienteles, and only for high-standard task variants, making broad roles costly exactly where quality standards bind. The narrower roles documented in the postings data are consistent with employers at the high end responding to this price. By contrast, establishments serving low-income consumers face weaker quality expectations and favor flexible staffing, which offers operational advantages of its own. In the worker-level analyses, much of the premium reflects worker selection rather than rent sharing alone, as one would expect if employers specialize jobs to deploy scarce worker skills.

The findings may seem to conflict with manufacturing-based evidence that enriched jobs can raise quality when embedded in broader systems of complementary work practices (Ichniowski, Shaw and Prennushi 1997; MacDuffie 1995). In both settings, task breadth comes at the cost of higher pay (Rahmandad and Ton 2020); yet manufacturing firms enrich jobs and pay that premium to deliver quality, while the firms studied here specialize to economize on it. A possible reconciliation, which this study does not test, is that how much quality depends on keeping tasks bundled differs across production settings. Where the steps of production are interdependent, as in craft work or low-buffer manufacturing, workers whose knowledge spans the process produce quality by catching problems that arise in one step and resolving them in others (MacDuffie 1995). In professional labor markets, adjacent domains of expertise likewise complement one another (Merluzzi and Phillips 2016). In both settings, unbundling to contain the price of breadth would sacrifice the quality the bundle produces. In the customer-facing services studied here, by contrast, tasks may unbundle more cleanly. At the high-end restaurant one manager oversees, silverware polishing "*happens in the back of the restaurant... away from guests*," and the manager avoids letting it "*take up the time*" of the staff who interact with guests, even as the establishment competes on "*higher touch service*" (Interviewee E). Unbundling may even benefit quality in this setting. Unlike goods, services are produced and consumed at once (Oliva and Sterman 2001), leaving no chance to correct an error before it reaches the customer, so each task must be executed flawlessly in real time, and a worker dedicated to a task may do so more reliably than one juggling many. Testing this account, and identifying more generally which features of production settings govern the relationship between quality competition and job breadth, is a promising direction for future research.

At the same time, limitations of the dataset and research design must be acknowledged. The sample is primarily composed of customer-facing service industries due to the characteristics of the SafeGraph dataset. These industries may differ from others in their employment and market strategies (Baum and Haveman 1997; Favaron, Di Stefano and Durand 2022; Osterman 2020), potentially limiting the generalizability of the results. The findings identify a novel mechanism of between-firm pay inequality; they do not establish that it operates universally across sectors.

I also use a job postings dataset to examine how differences in job design help explain the link between firms' product market strategies and pay levels. However, task descriptions in job postings may imperfectly capture the actual tasks performed on the job for various reasons. While it is reasonable to assume that posted task descriptions generally correlate with true job content on average, I cannot fully rule out the possibility that deviations between posted and actual tasks are systematically correlated with consumer income, which could bias the results.

A final limitation is that this study does not establish a causal relationship between consumer income and worker pay levels. A firm may adjust its jobs and pay to the consumers it draws, or it may redesign jobs and hire more skilled workers first and only then attract a richer clientele. Nor can the estimates fully rule out unobserved factors, beyond the consumer base, that move visitor income and pay together. What the design does rule out is the confound that has hindered efforts to connect worker pay to the consumers a firm serves: compared across areas, consumer income and wages correlate mechanically, as wealthier neighborhoods have both richer visitors and higher local wages and living costs. Comparing establishments within the same narrowly defined neighborhood removes this correlation, a control unavailable in earlier research (Batt et al. 2020), so the association that survives is more plausibly rooted in the product market. The within-establishment estimates additionally absorb all time-invariant establishment characteristics, though unobserved time-varying changes remain.

More fundamentally, the theory's claim survives either causal direction. The hypotheses state constraints tying pay and job design to the consumer base served, not a causal sequence. Whether establishments adapt jobs and pay to the consumers they draw, or enter with jobs and pay positioned for the consumers they expect, which positions are viable is set by the surrounding distribution of consumer income, and pay remains coupled to the consumer base. Centralized chains fit the same logic: uniform pay policies encode the segment a chain targets, and site selection matches outlets to the neighborhoods where that segment resides, so the account does not require site-by-site wage setting. What the evidence rejects is the third possibility, that the two are uncoupled, with high pay a discretionary strategy broadly available to firms regardless of whom they serve. The worker-selection results weigh against such free-standing discretion as well. The pay premium reflects who is employed rather than rent-sharing with a common pool of workers, as expected if job requirements are pinned down by the segment served rather than set freely by the employer. Future research can use natural experiments to establish causality in the relationship between consumer income and firm pay setting.

This research identifies an important but underexamined constraint on firms' pay decisions: consumer income. In contrast to perspectives that portray firm-level wage setting as relatively independent of market forces (Avent-Holt and Tomaskovic-Devey 2014; Ton 2014), the evidence presented here points to structured, rather than purely discretionary, variation in how firms respond to consumer demand – variation that is ultimately bounded by the pool of consumers available to them. At the same time, this perspective complements prior research on the economic constraints shaping firm pay setting (Berger et al. 2024; Card et al. 2018). Whereas this literature often abstracts from how employment practices are embedded within firms' broader strategic choices, the findings show that wage variation is systematically aligned with both the characteristics of consumer demand and firms' corresponding choices about job design.

For research on job design, the findings add a demand-driven source of task assignment. Prior research has shown how employers' efforts to minimize costs (Wilmers 2020), cope with uncertainty (Miner 1987; Tan 2015), and respond to technological change (Beane 2023) can generate variation in job design. Yet, how consumers contribute to the creation of different jobs has received little attention. Conversely, although scholars have increasingly examined how consumers and clients shape organizational practices and worker identities (DiBenigno 2022; Osterman 2020), their influence on task allocation remains underexplored. Linking task assignment to the consumer base fills that gap. Establishments serving wealthier consumers build jobs from higher-priced variants of similar tasks and define roles more narrowly, consistent with consumer expectations of quality entering job design through managers' efforts to deliver it.

Empirically, this is the first large-scale dataset to measure consumer income and worker pay at the establishment level, allowing me to address questions and potential confounders that prior research has been unable to examine directly. Previous firm-level studies have proxied the consumer base with prices or self-reported market segments rather than measuring consumer income (Osterman 2020; Talbert and Bose 1977). Studies that do measure consumer income aggregate to industry-by-region cells (Gottlieb et al. 2023; Wilmers 2017), where the role individual firms play cannot be observed. In contrast, this study measures both worker pay and consumer income at the establishment level on a large scale, making it possible to assess whether the income levels of the consumers an establishment serves are systematically associated with its workers' pay.

Theoretically, job specialization, rather than task richness, can be bundled with pay premiums to deliver higher-quality service reliably. Prior research in manufacturing has emphasized that broader task assignments can improve quality by fostering worker flexibility and cross-functional collaboration (Ichniowski, Shaw and Prennushi 1997; MacDuffie 1995). My findings suggest that this relationship operates differently in service settings. Firms serving higher-income consumers tend to assign workers narrower task sets, even though these consumers are likely to place greater emphasis on service quality. Task breadth is priced only at high-clientele firms, and only for high-standard task variants (Figure 7), suggesting that firms adopt specialized jobs to deliver quality where workers capable of performing multiple quality-producing tasks are scarce or costly. The relationship between job specialization and quality is thus context-dependent.

To practitioners, the findings speak directly to the good jobs debate. Advocates of high-road employment urge firms to pay well and invest in workers, treating these choices as available to any employer willing to make them (Ton 2014). The evidence here suggests that the room for such choices is unevenly distributed: it is tied to the consumers a firm serves, and employers whose customers cannot pay for quality have little margin to fund high wages. Exhortation alone will not create that margin. Yet the same evidence locates where the high road runs in customer-facing services. Establishments gained room to raise pay as their customer bases shifted toward higher incomes, and they delivered the quality those customers paid for through specialized, high-standard roles. For managers, the lever is market position and the job design that supports it, as much as pay policy itself. For advocates and policymakers, measures that raise incomes at the bottom of the consumer distribution may do double duty, expanding the demand for quality that sustains better-paying jobs.

More broadly, these findings underscore the need to integrate inequality among consumers into models of organizational design and labor market inequality. As firms position themselves to meet the preferences of different market segments, they restructure not only their products and services, but also their internal job architectures. In doing so, they help shape the broader structure of opportunity in a labor market increasingly organized around customer-facing service work. Understanding inequality, therefore, requires not just attention to institutional frictions or managerial discretion, but also a serious engagement with the ways consumers influence the design of work itself.

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A QCEW Blanket-Access States

Although the QCEW collects data from all states, some states do not provide unrestricted access to researchers. As a result, these states have been excluded from my analysis. Table A.1 details the states and territories included in this study.

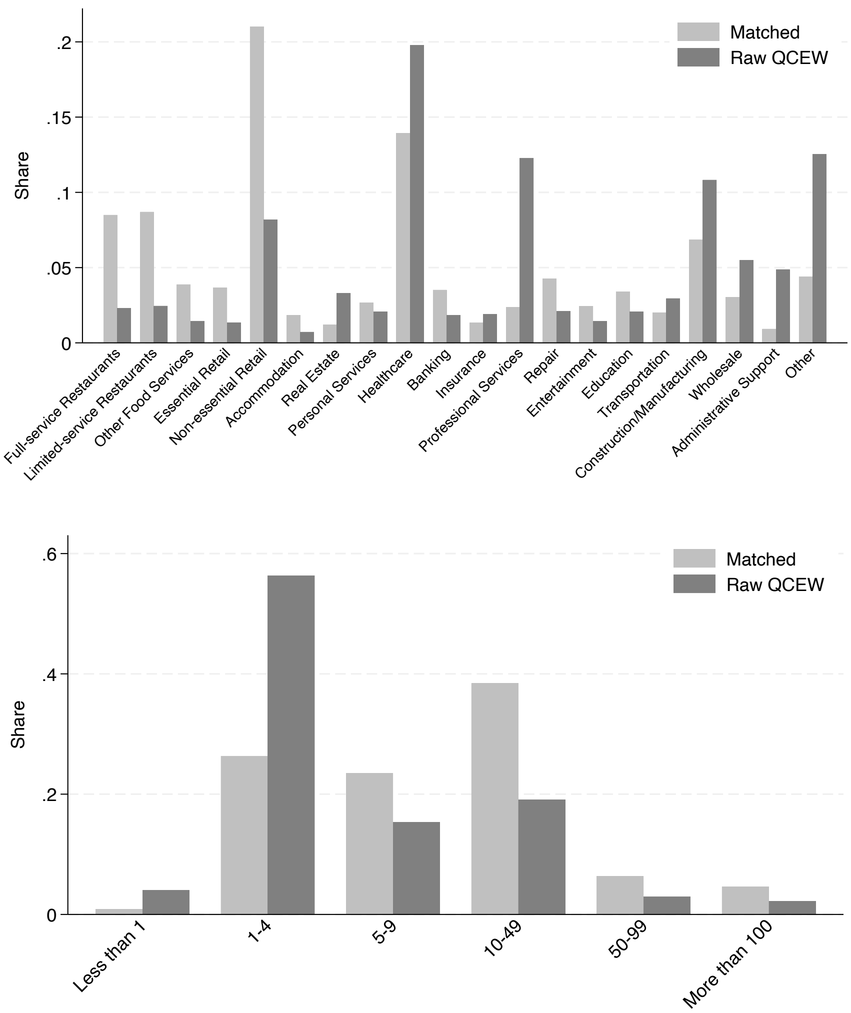
Table A.1: List of Included States and Territories

Alaska	Iowa	New Jersey	Texas
Alabama	Idaho	New Mexico	Utah
Arizona	Indiana	Nevada	Virginia
California	Kansas	Ohio	Virgin Islands
Connecticut	Maryland	Oklahoma	Vermont
Washington D.C.	Maine	South Carolina	Washington
Delaware	Minnesota	South Dakota	Wisconsin
Georgia	Missouri	Tennessee	West Virginia
Hawaii	Montana		

B Additional Details on SafeGraph-QCEW Match

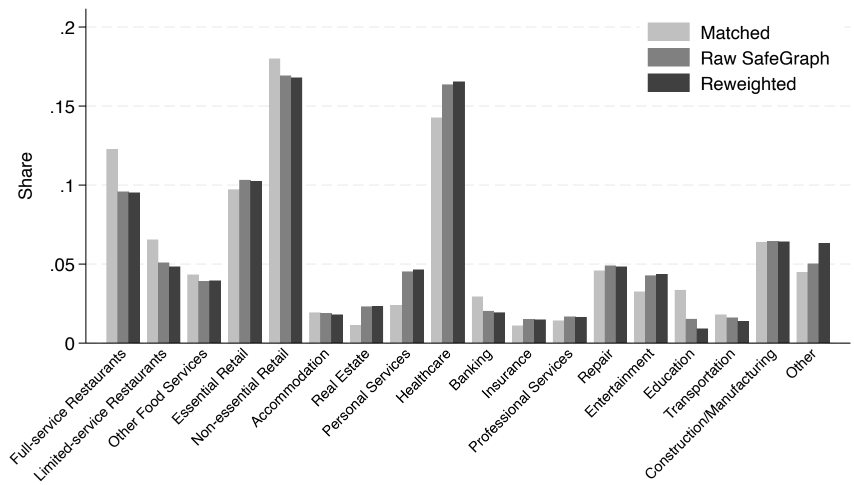
SafeGraph's Places dataset provides the names and addresses of points of interest (POIs), while QCEW collects establishment names and address information for U.S. businesses with more than three employees through the Annual Refiling Survey (ARS). To match these two datasets by establishment names and addresses, I used the user-written Stata package *reclink2* (Wasi and Flaaen 2015). Initially, I performed exact matches based on standardized workplace names and addresses, using pattern files included with the *reclink2* package. Following this, I conducted a series of fuzzy matches with the *reclink2* command, iteratively refining the process to minimize false positives and false negatives.

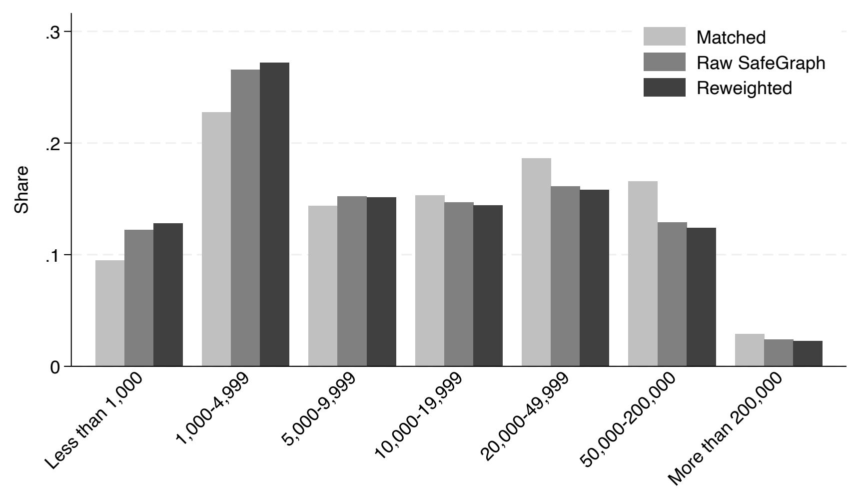
Figure A.1: Unmatched sample share by industry and number of quarterly visitors



The overall match rate is approximately 36% from the SafeGraph side and 20% from the QCEW side. The lower match rate from the QCEW side is expected, as many of the SafeGraph POIs are customer-facing businesses. However, even the SafeGraph match rate is not as high as desired, primarily due to missing addresses in QCEW. QCEW address information, sourced from ARS, excludes smaller businesses and only surveys one-third of its sampling frame annually. This aligns with the pattern in Figure A.1, which shows that smaller businesses, particularly those with fewer than five employees, are underrepresented in the matched dataset.

Figure A.2: Unmatched sample share by industry and number of quarterly visitors





On the other hand, as shown in Figure A.2, the disparity between matched and raw samples is less pronounced from the SafeGraph side.[7] This is somewhat reassuring given the low match rate, though there are still subtle differences between the matched and raw SafeGraph samples. Matched workplaces tend to attract more customers, with restaurants being overrepresented, while real estate, personal services, and healthcare are also underrepresented. To check whether these differences in size and industry distribution affect the main findings of the paper, I conduct a reweighting exercise described in Appendix C, which rebalances the matched sample toward the raw distribution and yields a similar estimate.

C Robustness Checks on the SafeGraph-QCEW Analysis

Table A.2: Robustness Checks on the Main Cross-sectional Pay Regression

	(1)	(2)	(3)	(4)	(5)
Visitor Income	0.064***	0.076***	0.054***	0.055***	0.074***
	(0.004)	(0.007)	(0.005)	(0.012)	(0.005)
Fixed effects:					
Time × NAICS × CBG	×	×	×	×	×
Controls	×	×	×	×	×
N	3,395,868	1,153,982	3,395,868	370,205	2,932,432

* Dependent variables for all models are logged average pay at establishment level from QCEW. Pay deflated with 2017 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. Column 1 is the main model from Table 2 Column (4). Column 2 only uses workplaces where more of their visitors are from neighborhoods with homogeneous income levels. Column 3 shows the weighted regression results that corrects for the changes in industry and visitor size shares from data merge. Column 4 shows the results from the analysis only using 2022 sample. Column 5 drops healthcare and manufacturing sectors. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

[7] Note that the industry distribution of the matched sample differs from Figure 2, as Figure A.2 represents unique places, while Figure 2 is at the place-by-time level.

Table A.2 presents the results from additional robustness checks for the main cross-sectional regression analysis in Column (1), which is equivalent to the results in Column (4) of Table 2. In Column (2), I use only those workplaces whose visitors are more likely from CBGs with homogeneous income levels. Since I approximate median visitor income across CBG-level averages from ACS, the value will become more accurate as the income levels within each CBG converge. I calculate establishment-level averages of within-CBG income variance across those CBGs where the visitors reside. Then, I use only those establishments with average within-CBG income variance lower than mean to replicate the model in Column (1). The coefficient increases as expected, suggesting possible measurement errors in visitor income driving attenuation bias.

Column (3) illustrates the results from a weighted regression, where I have reweighted each workplace by the ratio of the number of raw SafeGraph observations to matched sample observations in each industry by visitor number cell that the workplace is in. These cell definitions are consistent to what has been defined in Figure A.2. I try to address possible selection bias (see Appendix B for more discussion) coming from the merge process this way. Comparing the result with the main model in Column (1), while the coefficient decreases a little, a 10% increase in visitor income still predicts a statistically significant 0.5% increase in average worker pay, indicating that the main finding is not driven by the composition of the matched sample.

Column (4) only uses observations from 2022 to replicate the analysis in Column (1). While the SafeGraph dataset dates as far back as 2018, it is less reliable for the earlier periods as the number of mobile devices was smaller. On the other hand, visit patterns could have been contaminated by irregularities during the COVID-19 pandemic when there were frequent lockdowns or store closures. The 2022 data is free from both of these concerns. The result is again consistent with the main model, albeit the coefficient being a bit smaller.

Lastly, Column (5) drops the construction, manufacturing, and healthcare industries, which could be problematic for our analysis. While SafeGraph *visitors* have been assumed to represent *consumers* in the analyses, this assumption likely holds primarily for service industries. Additionally, wage rates in the healthcare sector are largely set by government and insurance companies (Osterman 2020) rather than at the establishment level. Including these industries may therefore bias the true correlation between consumer income and pay levels. Consistent with this expectation, Column (5) demonstrates a higher coefficient when these industries are excluded from the sample.

D Industry and Occupation Classifications

Figure 4 replicates the main cross-sectional pay regressions in Table 2, only using establishments from specific industries. The definition of industries is as below, following Massenkoff and Wilmers 2023.

- Full-service Restaurants: NAICS Code 722511
- Limited-service Restaurants: NAICS Code 722513
- Other Food Services: NAICS Codes 722000-722999
- Essential Retail: NAICS Codes 445000-447999
- Non-essential Retail: NAICS Codes 440000-459999
- Accommodation: NAICS Codes 721000-721999
- Real Estate: NAICS Codes 531000-531999
- Personal Services: NAICS Codes 812000-812999
- Healthcare: NAICS Codes 620000-629999
- Banking: NAICS Codes 522000-522999
- Insurance: NAICS Codes 524000-524999

- Professional Services: NAICS Codes 540000-549999
- Repair: NAICS Codes 811000-811999
- Entertainment: NAICS Codes 710000-719999
- Education: NAICS Codes 610000-619999
- Transportation: NAICS Codes 480000-499999
- Construction: NAICS Codes 230000-239999
- Manufacturing: NAICS Codes 310000-339999
- Wholesale: NAICS Codes 420000-429999
- Administrative Support: NAICS Codes 561000-561999

Figure 5 replicates the occupation-level pay regressions separately for each occupational group. The definition of the occupational groups is as below, with occupations outside these categories assigned to the residual group.

- Managerial: 2-digit SOC Code 11 (Management)
- Professional: 2-digit SOC Codes 13 (Business and Financial), 15 (Computer and Math), 17 (Engineering), 19 (Science), 23 (Legal)
- Frontline: 2-digit SOC Codes 31 (Healthcare Support), 35 (Food Services), 39 (Personal Services), 41 (Sales), 43 (Administrative Support), 47 (Construction and Extraction), 51 (Production), 53 (Transportation and Material Moving)

E OEWS match

The Occupational Employment and Wage Statistics (OEWS) dataset, a restricted-access resource from the Bureau of Labor Statistics, provides workplace-by-occupation employment and wage statistics collected through representative, repeated surveys. OEWS uses a three-year sampling strategy, where the establishments surveyed over a three-year period form a representative sample.

Under this design, an establishment sampled in year y is not surveyed again until $y + 3$. Each year, OEWS surveys about 400,000 establishments, resulting in a combined three-year sample of roughly 1,200,000 establishments.

I make two additional adjustments to the OEWS dataset. First, I create 5-digit occupation codes to harmonize the various 6-digit occupational classification systems used during the transition to the new Standard Occupational Classification (SOC) system in 2018. Second, I apply the methodology from Wilmers and Aeppli (2021) to convert the wage variable to a continuous scale, as most of the OEWS wage data was reported in intervals prior to 2020.

I merged the OEWS data with the SafeGraph-QCEW dataset using workplace-level identifiers common to both OEWS and QCEW. The merge process drops a large number of QCEW establishments as OEWS samples roughly 400,000 establishments annually, while QCEW covers the full universe of establishments across more than half the states. On the other hand, the state restriction of the QCEW sample leaves a significant portion of OEWS establishments unmatched (see Appendix A). I further drop manufacturing and healthcare industries from the matched sample used for analysis: the sampling strategy of OEWS is biased towards larger establishments with more workers, leading to the overrepresentation of these two problematic industries (see Appendix C for more discussion on why these could be problematic).

The final SafeGraph-QCEW-OEWS dataset is structured at the Quarter-by-Workplace-by-Occupation level, with a total of 2,130,753 observations and 286,565 Quarter-by-Establishment pairs. Table A.3 illustrates the descriptive statistics of this sample.

Table A.3: Descriptive Statistics of OEWS matched sample (N=2,130,753)

Variable	Mean	S.D.	Min.	Max.
log(occupational wage)	2.61	0.55	1.57	4.82
log(Median visitor income)	10.99	0.28	8.29	12.42
log(Average employment)	3.97	1.43	-1.09	10.67
log(Number of visitors)	8.25	1.85	1.81	14.29
Share Low wage Occ.	0.41		0	1
Share Mid. wage Occ.	0.30		0	1
Share High wage Occ.	0.28		0	1
Share Managerial Occ.	0.04		0	1
Share Frontline Occ.	0.69		0	1
Share Supervisor Occ.	0.06		0	1
Share Customer-facing Occ.	0.43		0	1

Note: Observations at Time × Workplace × Occupation level, 2018-2022. The sample drops healthcare and manufacturing industries.

F Additional Results from Occupational Analyses

First, I examine whether variations in occupational composition drive pay differences across establishments. To test this possibility, I estimate the relationship between establishment-level average pay and visitor income, both with and without controlling for occupational composition. To construct a measure of workplace-level occupational composition, I categorize occupations into ten groups based on their 2012 OEWS pay levels, accounting for their distribution across different workplaces. I assign pay group numbers to each five-digit SOC occupation, with 10 representing the highest-paid occupations and 1 the lowest. Empirically, I derive these decile groups by estimating occupation fixed effects from a two-way fixed effects model of wages on five-digit occupations and workplace identifiers, following Wilmers and Aeppli (2021). Using this approach, I compute the average occupational pay group number for each SafeGraph-QCEW-OEWS matched workplace, which serves as its occupational composition score.[8]

I estimate a series of workplace-level regressions using the SafeGraph-QCEW-OEWS matched sample, varying the granularity of regional fixed effects to account for the smaller sample size resulting from the OEWS coverage restriction.[9] If pay differences across establishments are primarily driven by differential composition of occupations, then controlling for the score should systematically attenuate the effect of visitor income on pay levels across different model specifications.

Table A.4: Cross-sectional Pay Regressions, Controlling for Occupational Composition

	(1)	(2)	(3)	(4)	(5)	(6)
Visitor Income	0.046***	0.040***	0.079	0.101*	0.130	0.126
	(0.012)	(0.012)	(0.055)	(0.054)	(0.115)	(0.115)
Occ. Composition Score		0.095***		0.068***		0.060***
		(0.001)		(0.006)		(0.011)
Fixed effects:						
Time × ZIP Code	×	×				
Time × NAICS	×	×				
Time × NAICS × ZIP Code			×	×		
Time × NAICS × CBG					×	×
Controls	×	×	×	×	×	×
N	222,778	222,778	14,213	14,213	4,007	4,007

* Observations at 2018-2022 workplace level, matching SafeGraph-QCEW-OEWS and dropping healthcare and manufacturing industries. Pay deflated with 2000 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. The Occupation Composition Score is the workplace-level average of globally calculated occupation pay deciles, which group occupations into 10 categories based on their 2012 average wages. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

The results, shown in Table A.4, provide limited evidence that occupational composition accounts for the relationship between visitor income and workplace pay. The modest decline in point estimates

[8] I use pay deciles for occupations instead of raw occupation fixed effects, as decile-based scores better capture the boundaries between occupations. This distinction is particularly important given that wages in the OEWS data are reported in intervals, which may cause fixed effects to imperfectly or biasedly reflect underlying wage differences.

[9] OEWS is based on a three-year rotating survey design, sampling approximately 400,000 establishments per year. More details are provided in Appendix E.

after controlling for occupation scores indicates that occupational differences explain only about 3% of the visitor income effect on pay in models with CBG fixed effects. Across various specifications comparing different samples, controlling for occupational composition score captures only a small portion, if any, of the visitor income effect on pay.

Motivated by this finding, I next estimate occupation-level regressions with and without occupation fixed effects. Including occupation fixed effects enables comparisons among workers in the same occupation but employed at establishments serving different consumer bases within the same neighborhood and industry. If higher consumer income predicts higher pay primarily through increased payoffs to each occupation – rather than through shifts in occupational composition – one would not expect a substantial reduction in the visitor income coefficient after including occupation fixed effects. The results in Figure A.3 show little reduction or a slight increase in the coefficient for visitor income, indicating that the positive association between consumer income and worker pay is largely driven by within-occupation pay differences rather than by changes in occupational composition across establishments.

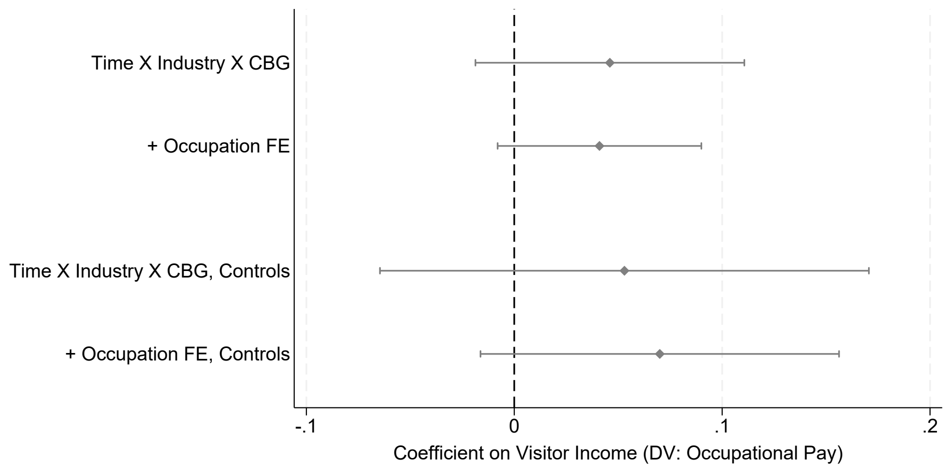
Table A.5 reports how these within-occupation estimates vary across occupational pay terciles, as described in Section 3.3.2; the association is concentrated among lower-paid occupations.

Table A.5: Pay Regressions on Visitor Income by Occupation Pay Quantiles

	(1)	(2)
Visitor Income	0.125**	0.209***
	(0.053)	(0.067)
Middle-paying Occs. × Visitor Income	0.010	-0.351***
	(0.062)	(0.103)
High-paying Occs. × Visitor Income	-0.124**	-0.104
	(0.064)	(0.099)
Fixed effects:		
Time × NAICS × CBG × Occ.	×	×
Common Controls	×	
Occs.-Specific Controls		×
N	22,333	22,333

* The regression model in Column (1) lets only the coefficients of visitor income vary for different occupational groups, whereas the model in Column (2) estimates separate coefficients for every variables including the controls. Observations at Time × Workplace × 5-digit Occupation level, matching SafeGraph-QCEW-OEWS 2018-2022. Dependent variables for all models are logged occupational wage predicted from OEWS via midpoint Pareto method following Wilmers and Aeppli 2021. Pay deflated with 2000 dollars. Visitor income is median visitor income predicted from SafeGraph with 2016-2020 combined ACS. Occupation pay quantiles defined from 2012 OEWS, fitting a two-way fixed effects (TWFE) model of pay with 5-digit occupations and workplaces. Time is at year-by-quarter-level. Controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Figure A.3: Visitor Income and Pay Within Occupations



Note: Observations at the 2018–2022 time × workplace × five-digit occupation level from the SafeGraph-QCEW-OEWS matched sample, dropping healthcare and manufacturing industries. The dependent variable is logged occupational wage predicted from OEWS via the midpoint Pareto method following Wilmers and Aeppli 2021; pay is deflated to 2000 dollars. Each estimate reports the coefficient on visitor income at the indicated level of fixed effects; controls include visitor shares of race, education, age, and sex, as well as the numbers of visitors and employees.

G Additional Details on Interviews

G.1 Sample

Table A.6: List of Interviewees

Name	Title	Restaurant	Market Segment
A	Assistant Manager	Willow Table	Casual
B	Assistant GM	Ember & Sage	Casual Fine Dining
C	Owner		
D	Owner	Everton Grill	Casual
		Hearth & Co.	Casual
E	Director of Operations	Stonewood	Casual Fine Dining
		Bellwether	Fine Dining
		Linden Café	Café
F	General Manager	Foxhall	Casual Fine Dining

Note: The names of the interviewees and the names of the restaurants are pseudonyms.

Table A.6 presents the sample of interviews conducted for this study. Potential interviewees were contacted through walk-ins and cold emails. In selecting establishments, I aimed to capture variation in price ranges and target customer segments (Trost 1986). Four of the five interviews were recorded and transcribed, with the exception of Interview D. The interviews took place between December 2024 and January 2025, each lasting approximately 30 minutes and conducted in person.

G.2 Interview Protocols

- How are roles structured within your establishment?
  - Are there certain roles that new employees typically do not take on?
  - How are tasks distributed within each role?
- What distinguishes job applicants who are hired from those who are not?
- How does your pay level compare to your competitors?

- What factors drive these differences?
  - Do these differences vary by role?
- How do employees transition between roles?
  - Does this happen within a single workday as well?
  - What is the reasoning behind these role transitions?
  - How do employees progress to supervisory positions?
- How significant is the difference between highly skilled workers and those with less experience?
  - Does this difference primarily stem from the hiring process or training?
  - What is the average tenure of employees in your establishment?
  - Do you offer internal training programs? How important are they to your business?
- What proportion of your workforce is part-time?
  - How do their job responsibilities compare to those of full-time employees?
  - How does their compensation differ from full-time employees?
  - Are there specific challenges associated with employing part-time workers compared to full-time staff?
- What aspects of the job make it attractive to employees?
- What are the biggest HR challenges you face?
- Are there differences in expectations for your workers compared to other restaurants?
  - What factors drive these differences?
- Do you have a specific target consumer base?
  - What do you think your customers value the most?
  - How do you tailor your operations to meet these demands?

H Additional Details on SafeGraph-Job postings Match

I match in SafeGraph visitor measures to the job postings dataset using employer names and MSA codes through *reclink2* module from Stata (Wasi and Flaaen 2015). I retain only those units that exceed a certain match score threshold in employer names. This threshold is determined by ensuring that at least 95% of sampled matches are correct, based on a manual review of 100 randomly sampled matches above the cutoff. The final matched sample includes 1,562,393 postings from 24,564 unique firm-by-metro-region units – representing roughly 5% of all SafeGraph establishments aggregated at the MSA level. From the job postings side, among firm-by-metro-region units in the Retail (NAICS 44 and 45) and the Accommodation and Food Services sector (NAICS 72) for which employer information is available, 30% are successfully matched to SafeGraph establishments. Note, however, that industry codes are available for fewer than 10% of firms in the Revelio Labs database, likely biasing the sample used in this match rate calculation toward larger, more publicly visible firms.

The modest match rate reflects three main limitations. First, not all firms post their jobs on Indeed. Second, because Revelio Labs relies on LinkedIn [10] pages for employer information – and LinkedIn primarily covers higher-skill segments of the labor market – the matched sample is skewed toward larger chains. In fact, employer names are missing for about half of the postings in the dataset. Third, unlike the QCEW, which reports both legal and trade names for each establishment, the Revelio Labs dataset includes only a company name, which may not always reflect the business name displayed at the establishment level (*e.g.*, on storefront signage).

[10] LinkedIn is an online platform for professional networking, where individuals create profiles to showcase their work experience, education, and skills – functioning as a dynamic, public-facing resumé. The platform also serves businesses, allowing companies to maintain organizational pages, post job openings, and recruit talent directly through LinkedIn's hiring tools.

I Additional Results on the SafeGraph-Job Postings Analyses

Table A.7 reports the full specialization estimates summarized in Figure 6 in the main text.

Table A.7: Number of Distinct Tasks and Consumer Income

	(1)	(2)	(3)	(4)	(5)	(6)
Visitor income (std.)	-0.0350***	-0.0351***	-0.0370***	-0.0690***	-0.0755***	-0.0777***
	(0.0095)	(0.0107)	(0.0109)	(0.0124)	(0.0140)	(0.0141)
Fixed effects:						
Qtr. × MSA × Ind. × Occ.	×	×	×	×	×	×
Controls	×	×	×	×	×	×
Sample	Full	Full	Full	NAICS 72	NAICS 72	NAICS 72
Task clusters	k=100	k=500	k=1000	k=100	k=500	k=1000
N	51,221	51,221	51,221	25,578	25,578	25,578

* The dependent variable is the logged number of distinct tasks at the indicated classification level, measured at firm × metro × quarter × occupation units. The sample drops healthcare and manufacturing industries. Visitor income standardized across firm × metro units. Controls include the logged number of visitors. Standard errors clustered by firm × metro in parentheses. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

In the main SafeGraph–job postings matched sample, I collapse multiple establishments belonging to the same firm at the metropolitan or micropolitan statistical area (MSA) level. While this approach facilitates matching, it may introduce bias if establishments within the same firm differ substantially in their consumer bases or employment practices. To address this concern, Table A.8 restricts the sample to firm × MSA units associated with a single establishment in the SafeGraph data – thus avoiding any aggregation. The specialization estimates are, if anything, larger than in the full sample, and the within-family variant estimate is likewise robust.

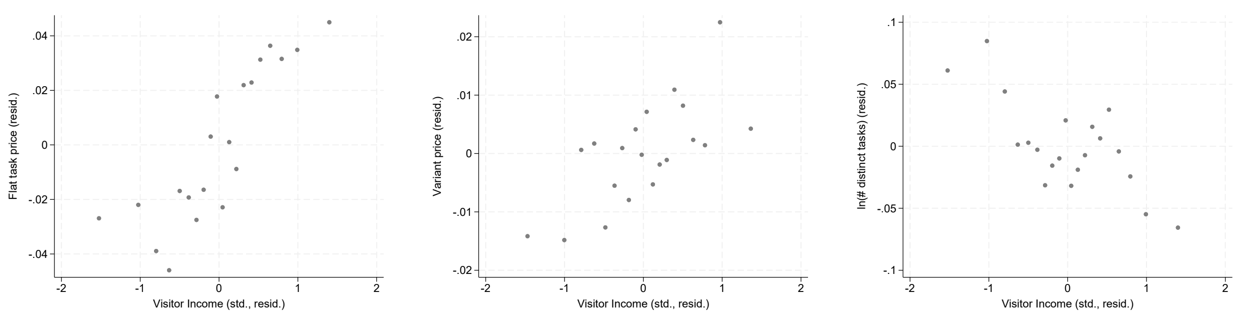
The distributional patterns shown in Figure A.4 provide further support for these results. Within estimation fixed-effect cells, consumer income exhibits an approximately monotonic relationship with the flat average task price, the within-family variant price, and the number of distinct tasks per job.

Table A.8: Single-Establishment Firms Only

	Flat average	Nested	Number of distinct tasks	
	(1)	(2)	(3)	(4)
Visitor income (std.)	0.0078	0.0137**	-0.0460***	-0.0487***
	(0.0129)	(0.0061)	(0.0143)	(0.0162)
Fixed effects:				
Qtr. × MSA × Ind. × Occ.	×	×	×	×
× Task family		×		
Controls	×	×	×	×
Task clusters	k=1000	k=1000-in-K=100	k=100	k=1000
N	13,686	42,428	13,686	13,686

* The sample is restricted to firm × metro units operating exactly one SafeGraph establishment in the metro area, so that each unit approximates a single establishment. Estimates are at firm × metro × quarter × occupation units, and in column (2) at task-family cells within those units. The sample drops healthcare and manufacturing industries. Visitor income standardized across firm × metro units. Controls include the logged number of visitors. Task prices from the full-universe 2022 salary regression. Standard errors clustered by firm × metro in parentheses. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Figure A.4: Binned Scatterplots of Job Design Outcomes and Visitor Income, Within Estimation Fixed-Effect Cells



(a) Flat average task price (b) Within-family variant price (c) Number of distinct tasks

* Data from SafeGraph matched with Revelio job postings, 2022, dropping healthcare and manufacturing industries. Each panel plots residualized bin means of the outcome against residualized visitor income within the fixed-effect cells of the corresponding estimation. Panels (a) and (c) residualize within quarter × MSA × industry × occupation cells, with observations at firm × metro × quarter × occupation units; panel (b) additionally interacts these cells with task families from the k=1000-in-K=100 nesting, with observations at task-family cells within those units. Visitor income predicted with 2016-2020 combined ACS.

J Classification of Task Families

Table A.9 lists all $K = 100$ task families with their customer-facing and frontline classifications. Customer-facing status reflects the substantive content of each family – whether performing its tasks typically involves direct interaction with the firm's customers, guests, clients, or patients. The classification was generated with a large language model (Claude Fable 5, Anthropic) from each family's label and most frequent member tasks, blind to the estimation results, with borderline families reviewed manually. I use this content-based approach rather than a keyword rule, such as the presence of "customer," "guest," or "client" in task descriptions, because keyword matching misfiles core service families, including food service and cashier work, as non-customer-facing. Frontline status is assigned empirically rather than judgmentally: a family is classified as frontline when its occurrences are disproportionately concentrated in frontline occupations, as defined in Appendix D. Concretely, a family counts as frontline when the share of its occurrences appearing in frontline occupations exceeds the sample-wide share of 76.8 percent, which flags 31 of the 100 families.

The most consequential borderline cases are supervisory families that combine customer contact with team oversight (for example, "Oversee retail operations, customer service, and inventory" is classified as customer-facing, while "Oversee hospitality and retail operations" is not). The customer-facing differential is sensitive to these calls, whereas the frontline split is insensitive to them – one reason the frontline concentration is the sharper and more robust of the two patterns in Table 6.

Table A.9: Classification of Task Families (K = 100)

Task family	CF	FL
Administer insurance claims and financial services		
Administer real estate leasing and tenant management	✓	
Analyze compliance data and manage regulatory risks		
Analyze data to inform business decisions		
Analyze financial data and generate reports		
Analyze financial data for compliance insights		
Assist with personal care in home settings	✓	
Conduct maintenance, safety operations on equipment		
Conduct medical diagnostics and recordkeeping	✓	
Conduct safety inspections and maintenance operations		
Conduct safety inspections, facility maintenance, and compliance		
Coordinate behavioral health and therapeutic services	✓	
Coordinate construction project management and compliance		
Coordinate educational and outreach initiatives in academic and community settings	✓	
Coordinate healthcare administrative tasks	✓	✓
Coordinate interdisciplinary clinical care in healthcare	✓	
Coordinate program managment		
Coordinate resident care in healthcare facilities	✓	
Coordinate vehicle loading, delivery, and customer service	✓	
Deliver comprehensive nursing care	✓	
Deliver nursing care and health education	✓	
Design, deliver, and assess educational programs	✓	
Develop and implement marketing and digital content		
Develop and integrate software and system solutions		
Direct engineering projects and provide technical expertise		
Direct retail banking financial operations		✓
Enforce health and safety standards in various settings		✓
Enforce infection control protocols in healthcare		
Enhance client relations and business growth	✓	
Enhance guest experience in hospitality services	✓	✓
Enhance quality and safety in operations		
Enhance sales and customer relationship management	✓	
Enhance team performance in healthcare and retail	✓	
Execute technical and manual tasks in production settings		✓
Facilitate educational environments	✓	
Facilitate professional growth and compliance in healthcare and corporate environments		
Food preparation and service in hospitality	✓	✓
Implement patient-centered care plans	✓	
Lead quality and process improvement in manufacturing		
Lead retail team development and customer engagement	✓	✓
Lead team innovation and problem-solving initiatives		
Lead technology solution development and integration		
Lead workforce development and team leadership initiatives		
Maintain accurate healthcare documentation		
Maintain and troubleshoot IT systems for operational integrity		
Maintain cleanliness and food preparation in hospitality		✓
Maintain documentation compliance		✓
Maintain facility cleanliness, safety, and equipment		✓
Maintain operational standards and process inventory		
Manage client documentation for regulatory compliance		
Manage data entry and record confidentiality		
Manage emergency responses and security incidents		
Manage financial reporting and contract negotiations		
Manage healthcare compliance and audits		
Manage IT security and incident response operations		
Manage medication administration and health records		
Manage nursing operations in healthcare settings	✓	
Manage patient care coordination	✓	
Manage pharmacy patient care services	✓	
Manage production quality and streamline manufacturing efficiency		✓
Manage property upkeep, maintenance, and operational efficiency		
Manage retail financial transactions and cash handling	✓	✓
Manage retail inventory, merchandising, and stock organization		✓
Manage retail operations and customer engagement	✓	
Manage staff training and performance evaluations		
Manage team compliance with operational standards		
Operate and maintain heavy machinery for safety and efficiency		
Operate equipment and perform maintenance in warehouses		✓
Optimize retail sales, merchandising, and customer service	✓	✓
Optimize team operations and performance management		
Oversee construction compliance and project execution		
Oversee employee recruitment, training, and performance		
Oversee hospitality and retail operations		
Oversee production quality and operations management		✓
Oversee retail inventory management and order processing		✓
Oversee retail operations, customer service, and inventory management	✓	✓
Oversee warehouse operations, inventory management, and logistics		✓
Perform clerical tasks		✓
Perform housekeeping and organizing in residential settings		✓
Perform quality assurance and testing in various industries		✓

Table A.9: Classification of Task Families (K = 100), continued

Task family	CF	FL	Task family	CF	FL
Perform scientific research and report findings			Provide IT support and manage technical solutions		
Process and analyze medical specimens in lab		✓	Provide technical solutions and process optimization		
Provide automotive repair and maintenance services			Resolve customer issues and manage vendor relations in retail	✓	
Provide banking financial services and customer support	✓		Service customers in retail	✓	✓
Provide client support and case management services	✓		Sterilize medical equipment, maintain safety standards		
Provide client-centric social support services	✓	✓	Supervise policy compliance and team performance		✓
Provide clinical patient care in healthcare	✓		Supervise staff compliance and development in HR		
Provide customer service and administrative support	✓	✓	Support daily patient care and healthcare operations	✓	
Provide customer service in hospitality settings	✓	✓	Troubleshoot and repair technical equipment		✓
Provide exceptional retail customer service	✓	✓	Uphold sanitation and safety in workspaces		✓

* CF marks customer-facing families, whose tasks typically involve direct interaction with the firm's customers, guests, clients, or patients, coded with a large language model from each family's label and most frequent member tasks. FL marks frontline families, whose occurrences are disproportionately concentrated in frontline occupations as defined in Appendix D. Borderline customer-facing calls are discussed in the text.

K Glassdoor match

Glassdoor is an online platform where users voluntarily share pay information for their current or past jobs, along with short text reviews. For this study, I utilize only the pay-related data from Glassdoor and match it with the SafeGraph visitor income variable. The Glassdoor dataset includes various compensation components, such as base salary, cash and stock bonuses, profit sharing, sales commissions, and tips. For the worker-selection analysis in Section 4.3.3, I consolidate these components into a single logged hourly pay variable; the pay-component analyses in Appendix L examine base and extra pay separately.

The Glassdoor pay dataset spans from 2006 to 2023, containing over 21 million pay records submitted by individual users, with approximately 40% of observations coming from 2021 onward. However, unlike the SafeGraph dataset, Glassdoor provides regional information only at the city level, and employer data is recorded at the firm level rather than the establishment level. To align the datasets, I aggregate visitor information from SafeGraph to the firm-by-city level and match it with employer information from individual pay reviews in Glassdoor. Specifically, I use city, industry, and employer name variables to conduct a fuzzy match using the Stata module *reclink2* (Wasi and Flaaen 2015).

To fully utilize Glassdoor's extended time range, I compute the average consumer income quantile for each firm-by-city in the SafeGraph dataset. Specifically, I first calculate each firm-by-city unit's visitor-income decile while controlling for time, industry, and city, then average these quantiles within each unit over time. I then define a binary indicator for a high-income consumer base, assigning a value of 1 if the average decile value exceeds 5.5. I use this coarse binary measure because visitor income is observed for 2018–2022 while the pay reviews span 2006–2023, so applying the measure to other years assumes that units maintain a stable consumer base over that period. Whether a unit serves a high- or low-income clientele is more plausibly stable over time than its exact income level, making the binary classification more reliable for this extension.

After the initial matching between Glassdoor and SafeGraph, I retain only those firm-by-city units that exceed a certain match score threshold. This threshold is determined by ensuring that at least 95% of sampled matches are correct, based on a random sample of 100 observations above the threshold.

Table A.10: Descriptive Statistics of Matched SG-Glassdoor Sample (N=1,924,369)

Variable	Mean	S.D.	Min.	Max.
log(Hourly pay)	2.94	0.66	1.00	6.00
log(Base pay), reporters	2.85	0.57	1.64	4.37
log(Extra pay), reporters	1.06	2.41	-7.64	5.33
log(Number of visitors)	8.34	1.92	1.81	14.12
log(Number of employees)	9.45	2.62	0	15.42
Years of relevant experience	4.20	5.44	0	60
High-income consumers	0.59	0.49	0	1

Note: Each observation represents a pay review submitted by users in Glassdoor and is linked to SafeGraph visitor information at establishment level. The sample drops healthcare and manufacturing industries. Pay includes both base and extra pay. Hourly pay, base pay, and extra pay are winsorized at the 1st and 99th percentiles. The dummy variable for high-income consumer base and the number of visitors from SafeGraph averaged across time within each establishment. Number of employees recorded at firm level from Glassdoor.

The final matched dataset comprises 2,241,750 pay reviews from 232,410 firm-by-city units spanning the years 2006–2023. As in the OEWS and job-postings analyses (Appendix E; Section 4.1), I drop healthcare and manufacturing industries, leaving 1,924,369 reviews from 223,025 firm-by-city units for the analyses. Table A.10 presents the descriptive statistics for the key variables used in the analyses.

L SafeGraph-Glassdoor Analysis

In Glassdoor, users indicate whether their compensation is reported on an hourly, monthly, or annual basis and separately disclose pay components such as base pay, bonuses, tips, and profit sharing. This reporting structure allows me to assess whether differences in work hours or pay composition underlie the correlation documented in Section 3.3.1. Specifically, I first examine the relationship between consumer income and worker pay using hourly pay measure derived from Glassdoor. I then assess whether this relationship is driven by base pay or by any non-base pay components. In the latter analysis, I restrict the sample to workers who report extra pay, as these users constitute a smaller subsample and may differ systematically in reporting behavior.

Table A.11: Consumer Income and Worker Pay: Hourly Pay, Base Pay, and Extra Pay

Sample	All	With Extra Pay		
Dependent Variable	Total	Total	Base Only	Extra Only
High-inc. Consumer	0.0428***	0.0221***	0.0317***	-0.0150
	(0.0052)	(0.0085)	(0.0111)	(0.0217)
Fixed effects:				
Yr. × City × NAICS	×	×	×	×
Controls	×	×	×	×
N	1,730,431	376,621	375,910	376,621

* Observations at year by worker level, 2006-2023, matching SafeGraph visitor income data to Glassdoor pay reviews data. The sample drops healthcare and manufacturing industries. I replicate the analysis using the full sample and the subsample of workers who report extra pay. All dependent variables are measured in hourly terms and logged, using three alternative pay measures. *Total* includes base pay and additional compensation, including salaries, wages, bonuses, and tips. *Base only* includes salaries and wages, excluding bonuses and tips. *Extra only* includes bonuses and tips only. The dummy variable for high-income consumer base is defined at firm level, controlling for Time × City × NAICS cells. Controls include logged visitor counts, logged firm size, and employment status. Standard errors clustered by employer × city in parentheses. * Statistically significant at the .10 level; ** at the .05 level; *** at the .01 level.

Table A.11 decomposes the premium by pay component. The estimates indicate that high-income consumers are consistently associated with higher worker pay even after accounting for potential differences in work hours, and that base pay accounts for most of this relationship, with no significant association for extra pay (bonuses and tips). Taken together, these patterns imply that the findings in Section 3.3.1 are unlikely to be driven solely by differences in work hours or by tips.

Next, I examine whether worker selection can account for this relationship by controlling for unobserved, time-invariant worker characteristics. To do so, the sample used for Table 7 is restricted to users who submitted multiple job reviews at matched firm-city units, leaving 143,434 reviews, roughly eight percent of the matched sample. The results indicate that worker fixed effects absorb most of the pay variation previously attributed to consumer income, suggesting that worker selection plays a central role in explaining pay differences across firms serving consumers with different income levels.